

The Unreasonable Effectiveness of Mathematics in Physics

HiSEP Seminar 2026

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In 1960, Eugene Wigner marveled that mathematics is **unreasonably effective**

COMMUNICATIONS ON PURE AND APPLIED MATHEMATICS, VOL. XIII, 001–14 (1960)

The Unreasonable Effectiveness of Mathematics in the Natural Sciences

Richard Courant Lecture in Mathematical Sciences delivered at New York University,
May 11, 1959

EUGENE P. WIGNER

Princeton University

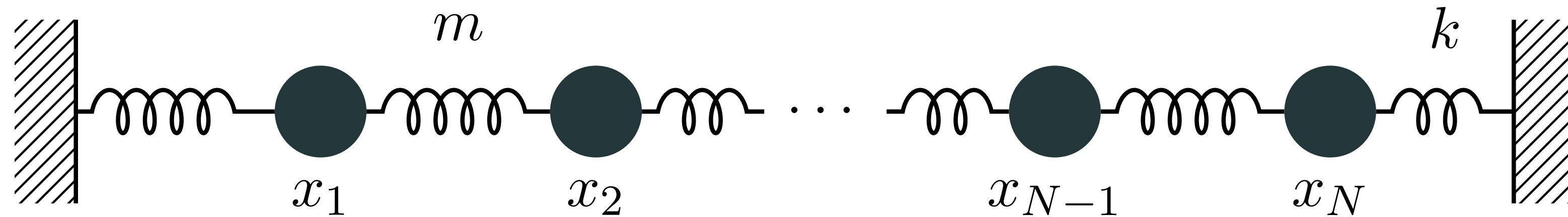
*“and it is probable that there is some secret here
which remains to be discovered.” (C. S. Peirce)*

Let me end on a more cheerful note. The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve. We should be grateful for it and hope that it will remain valid in future research and that it will extend, for better or for worse, to our pleasure even though perhaps also to our bafflement, to wide branches of learning.

SECTION 1

Classical Mechanics

The system: N masses coupled by springs



Newton's law for the displacement $x_j(t)$ of mass j (walls: $x_0 = x_{N+1} = 0$):

$$m \ddot{x}_j = k(x_{j+1} - x_j) - k(x_j - x_{j-1}) = -k(2x_j - x_{j+1} - x_{j-1})$$

Write it as one matrix equation by a vector $\mathbf{x} = (x_1, \dots, x_N)^\top$:

$$\ddot{\mathbf{x}} = -\frac{k}{m} A \mathbf{x}, \quad A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{pmatrix}$$

- The problem reduces to finding *eigenvectors* $\mathbf{v}$ and *eigenvalues* λ :

$$A\mathbf{v} = \lambda\mathbf{v}$$

- **Assumption:** a sampled sine wave, $v_j = \sin(j\varphi)$.

- Using $\sin((j+1)\varphi) + \sin((j-1)\varphi) = 2\cos\varphi \sin(j\varphi)$:

$$(A\mathbf{v})_j = 2\sin(j\varphi) - \sin((j+1)\varphi) - \sin((j-1)\varphi) = (2 - 2\cos\varphi) \sin(j\varphi)$$

- **Imposing boundary condition:**

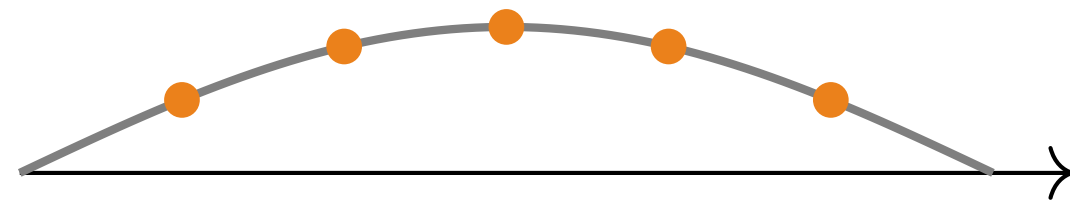
$$- v_0 = \sin 0 = 0: \text{automatic.}$$

$$- v_{N+1} = \sin((N+1)\varphi) \stackrel{!}{=} 0 \implies \varphi_a = \frac{\pi a}{N+1}, \quad a = 1, \dots, N.$$

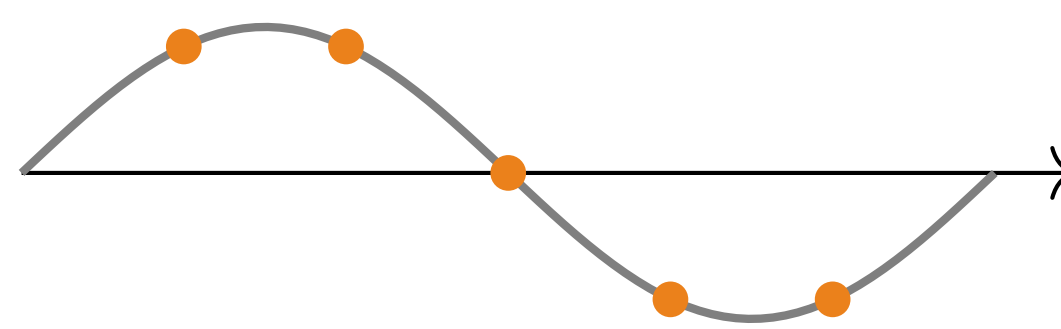
- **Result:**

$$v_j^{(a)} = \sin \frac{\pi a j}{N+1}, \quad \lambda_a = 4 \sin^2 \frac{\pi a}{2(N+1)}.$$

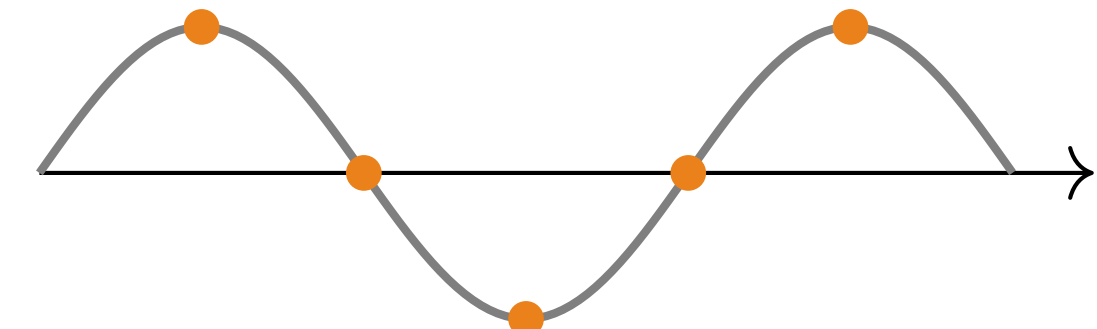
Each **normal mode** oscillates rigidly, with frequency $\omega_a = \sqrt{\frac{k}{m}} \lambda_a = 2\sqrt{\frac{k}{m}} \sin \frac{\pi a}{2(N+1)}$:



mode $a = 1$ (slowest)



mode $a = 2$



mode $a = 3$

- The general motion is a *superposition* of these modes — and writing a motion in this basis is exactly a **discrete Fourier (sine) transformation**:

$$x_j(t) = \sum_{a=1}^N c_a \sin \frac{\pi a j}{N+1} \cos(\omega_a t + \varphi_a)$$

- For $N \rightarrow \infty$ the chain becomes a string: the equation becomes the **wave equation**

$$\frac{\partial^2 x}{\partial t^2} = v^2 \frac{\partial^2 x}{\partial s^2}$$

and the sum over modes becomes the **Fourier series**.



Fourier

KÖRNER FOURIER ANALYSIS

FOREWORD BY TERENCE TAO

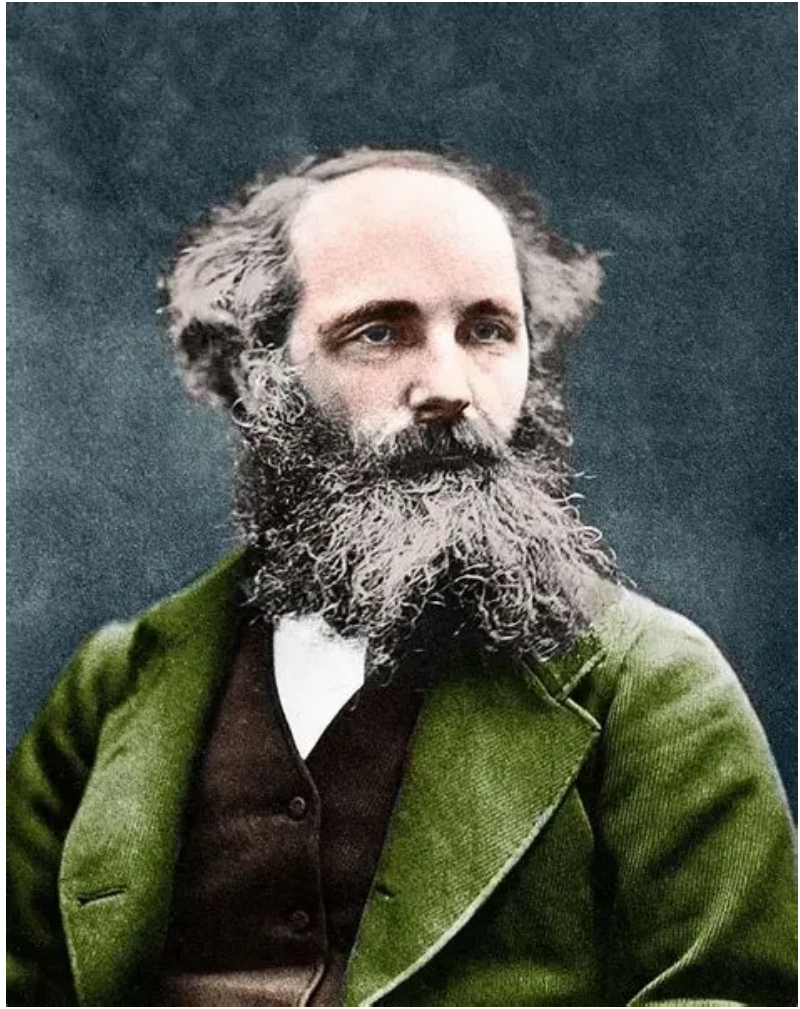


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SECTION 2

Electromagnetism

All of electricity, magnetism, and optics in four lines:



Maxwell

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

(Gauss)

$$\nabla \cdot \vec{B} = 0$$

(Gauss for magnetism)

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

(Faraday)

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

(Ampère–Maxwell)



Heaviside

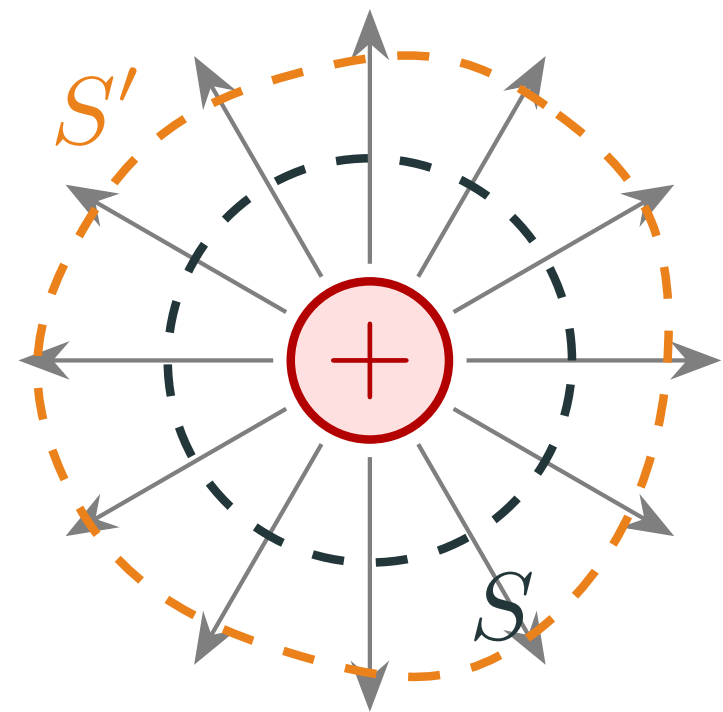
- In empty space they combine into a **wave equation** with speed

$$\frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m/s} \quad \implies \quad \text{light is an electromagnetic wave.}$$

- The equations fix this speed with no reference to any observer — the seed of relativity (Part 4).

- **Stokes' theorem** says: $\int_M d\omega = \int_{\partial M} \omega$
- Applied to Gauss's law $\nabla \cdot \vec{E} = \rho/\epsilon_0$:

$$\int_V \nabla \cdot \vec{E} dV = \oint_{S=\partial V} \vec{E} \cdot d\vec{S} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

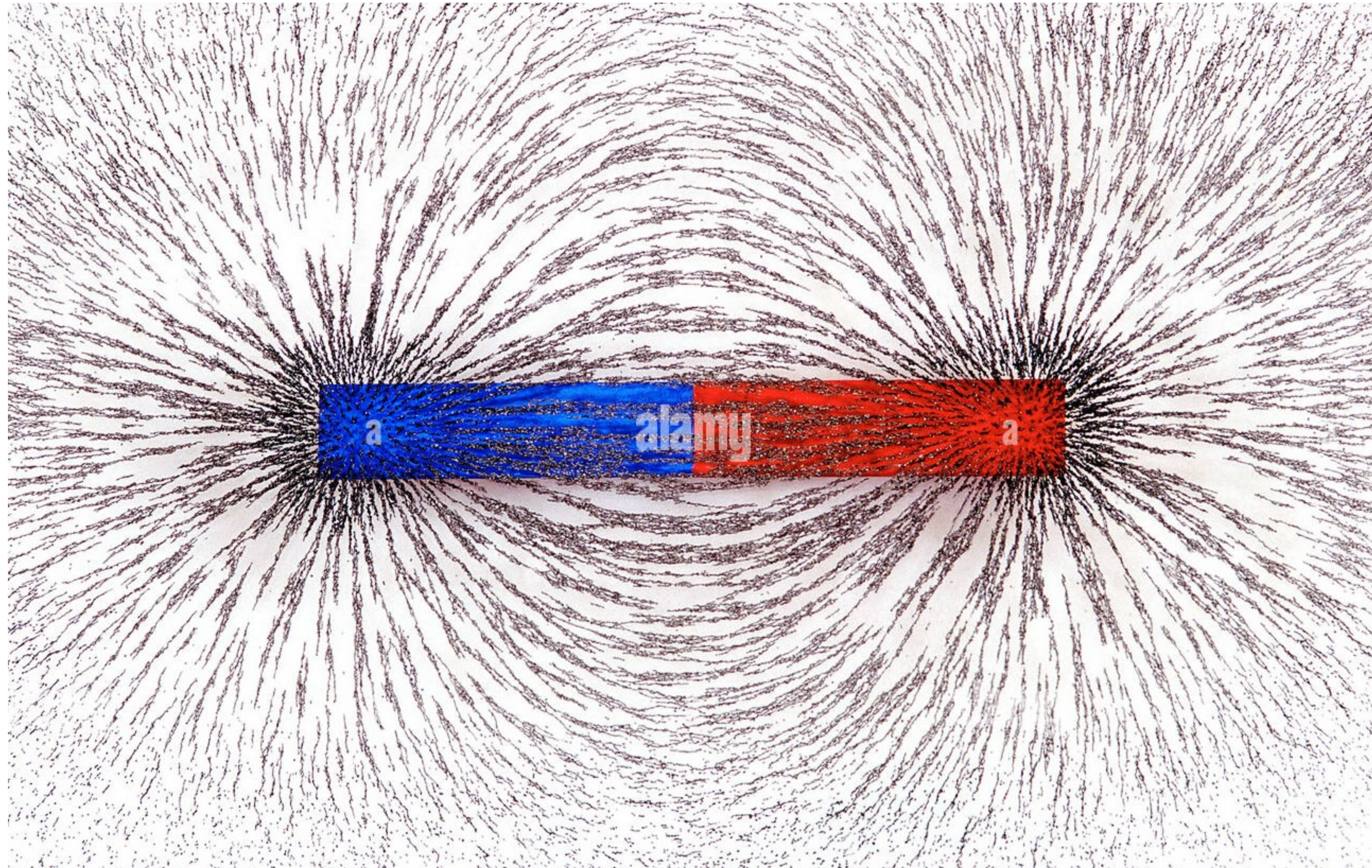


$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

positive charge — source

- **The shape of the volume V does not matter:** deforming $S = \partial V$ into S' crosses no charge, so the flux is unchanged — **only the charge inside counts.**

How about magnetic field $\vec{B}$?



$$\nabla \cdot \vec{B} = 0$$

- $\nabla \cdot \vec{B} = 0$ says magnetic field lines never begin or end: **there are no magnetic charges**.
- For any vector field $\vec{A}$, the identity $\nabla \cdot (\nabla \times \vec{A}) = 0$ holds. So writing $\vec{B} = \nabla \times \vec{A}$ solves the equation *automatically* — and on $\mathbb{R}^3$ the converse is also true:

$$\vec{B} = \nabla \times \vec{A} \iff \nabla \cdot \vec{B} = 0 \quad (\text{on all of } \mathbb{R}^3).$$

- $\vec{A}$ is the **vector potential**; it is not unique: $\vec{A} \rightarrow \vec{A} + \nabla \chi$ gives the same $\vec{B}$ (**gauge freedom**).
- Is $\vec{A}$ real physics, or just bookkeeping?

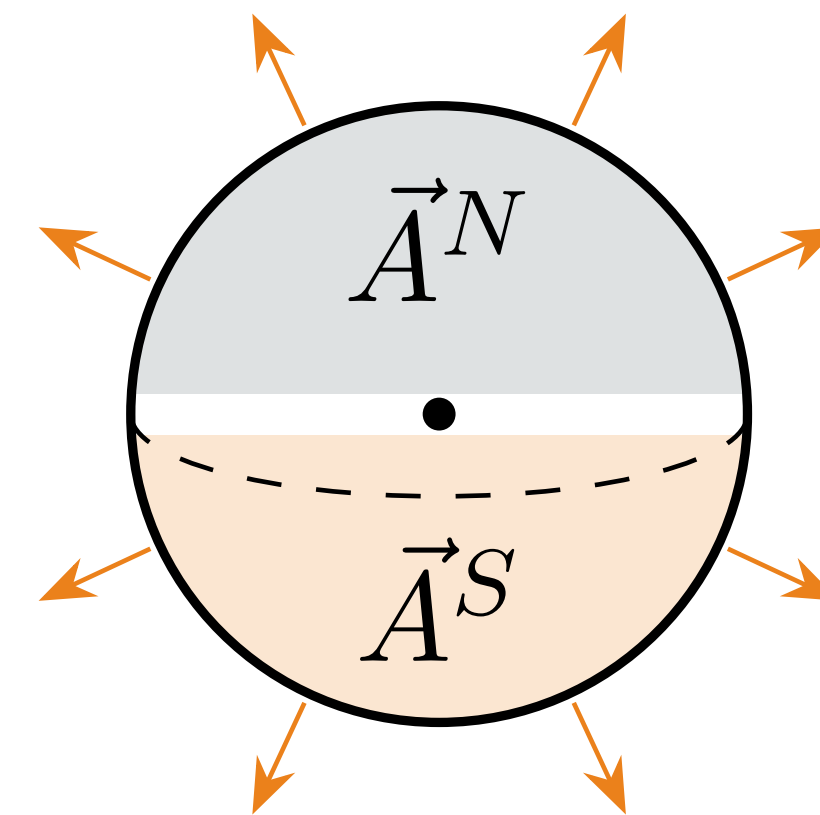
- *Imagine* a magnetic charge: a radial $\vec{B}$ with total flux $\oint_{S^2} \vec{B} \cdot d\vec{S} = g \neq 0$ through a sphere.
- But if $\vec{B} = \nabla \times \vec{A}$ everywhere, this flux must be zero: the hole at the origin changes the rules — topology enters.
- **Dirac's monopole (1931):** two potentials, one per hemisphere,

$$\vec{A}^{N/S} = \pm \frac{g}{4\pi r} \frac{1 \mp \cos \theta}{\sin \theta} \hat{\varphi},$$

glued on the overlap by a *gauge transformation*
 $\chi = g\varphi/2\pi$.



P. A. M. Dirac



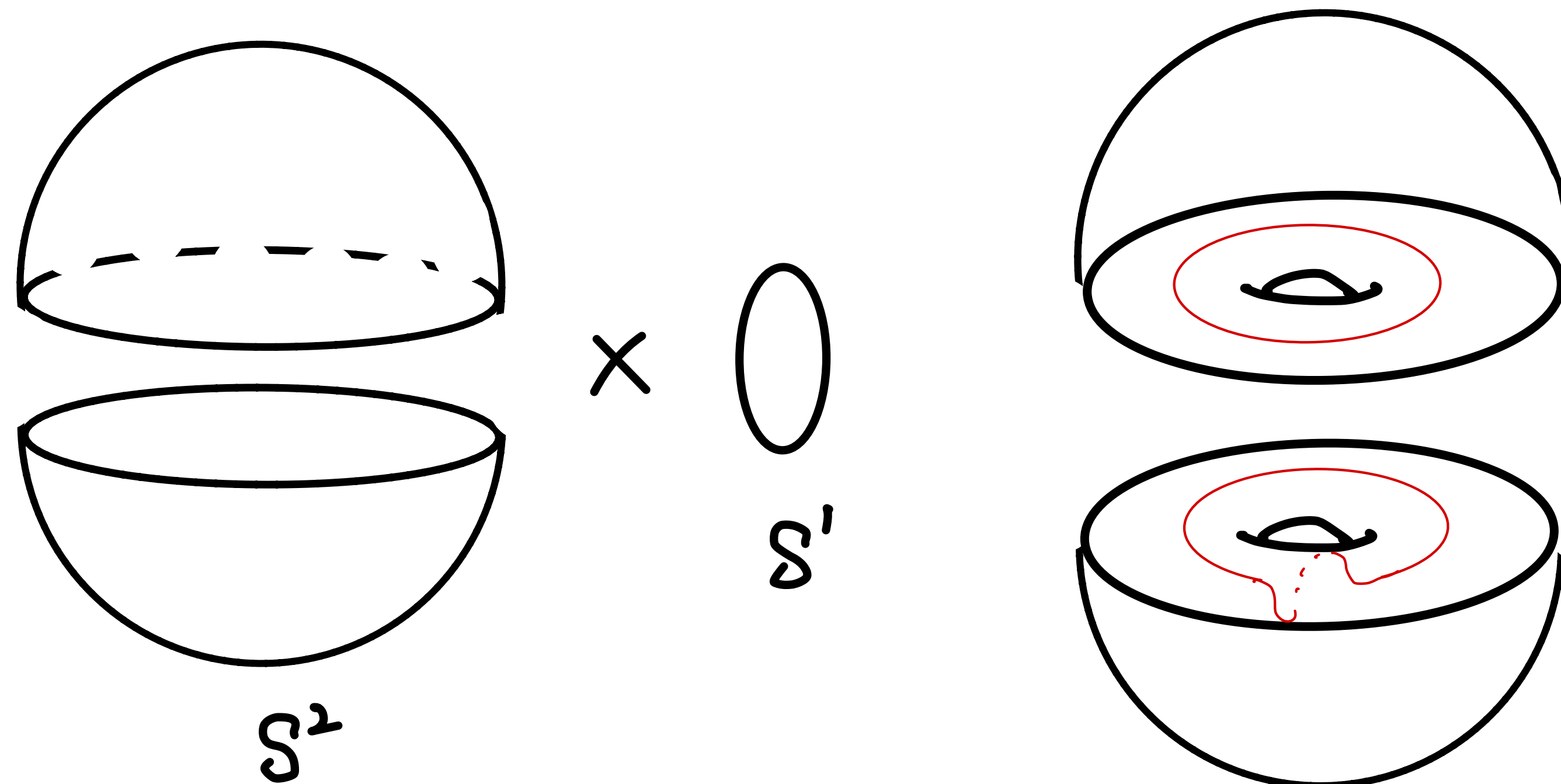
flux g through the sphere

Dirac's two glued potentials form a $U(1)$ **bundle over** S^2 — circles of phases glued with a *twist*.

- The simplest ($n = 1$) is the **Hopf bundle** $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$: for $(z, w) \in \mathbb{C}^2$ with $|z|^2 + |w|^2 = 1$,

$$\pi(z, w) = (|z|^2 - |w|^2, 2\operatorname{Re}(z\bar{w}), 2\operatorname{Im}(z\bar{w})) \in S^2.$$

- The total space is S^3 , **not** $S^2 \times S^1$: the fibers are linked.



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- The total space is S^3 , **not** $S^2 \times S^1$: the fibers are linked.
- The twist is measured by the **first Chern number** (Chern 1946) of $F = dA$: $c_1 = \frac{1}{2\pi} \int_{S^2} F = n \in \mathbb{Z}$ (the monopole charge).

Charge is quantized because the Chern number is an integer — and π will soon reappear as the *Bloch sphere* of a spin.



S.-S. Chern

Gauss's linking integral

A current I around a loop C_2 creates a magnetic field (**Biot–Savart** 1820):

$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \oint_{C_2} \frac{d\vec{r}_2 \times (\vec{r} - \vec{r}_2)}{|\vec{r} - \vec{r}_2|^3}.$$

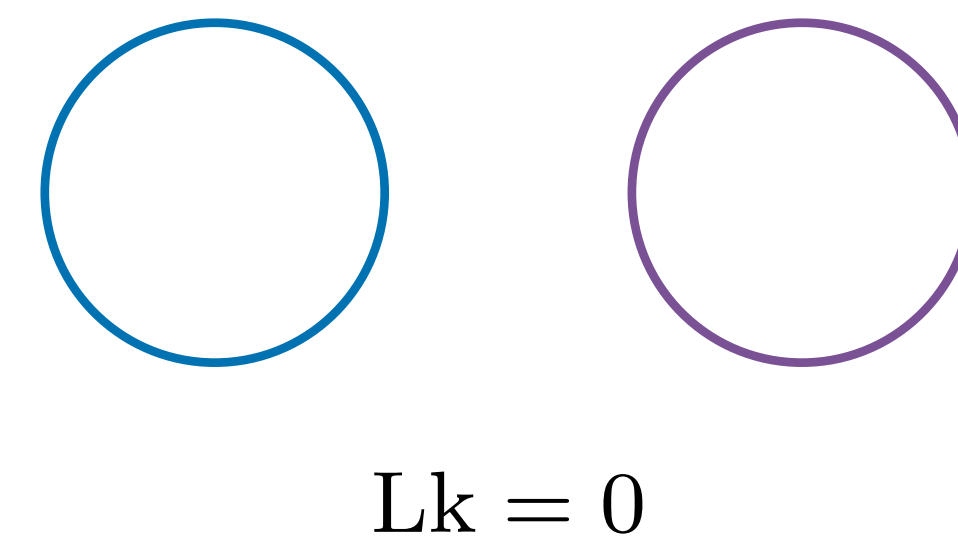
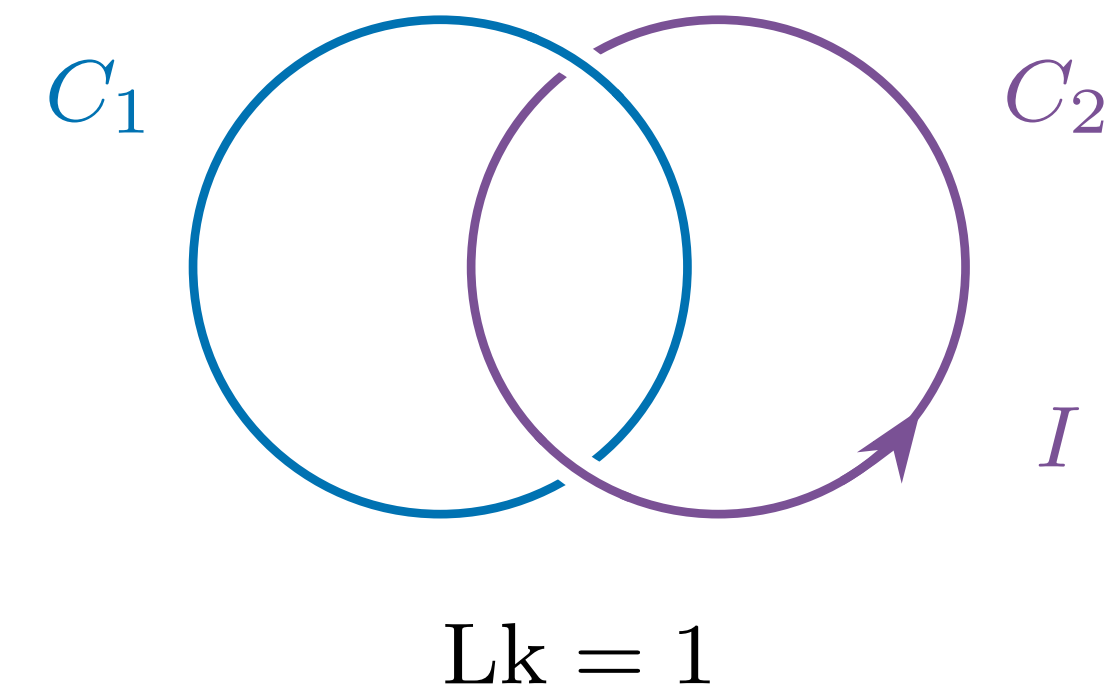
Ampère's law: circulating $\vec{B}$ around a second loop C_1 counts how often the current threads it:

$$\oint_{C_1} \vec{B} \cdot d\vec{r}_1 = \mu_0 I \times (\# \text{ times } C_2 \text{ threads } C_1).$$

Combining the two, the physics ($\mu_0 I$) cancels — what remains is pure geometry:

$$\text{Lk}(C_1, C_2) = \frac{1}{4\pi} \oint_{C_1} \oint_{C_2} \frac{(d\vec{r}_1 \times d\vec{r}_2) \cdot (\vec{r}_1 - \vec{r}_2)}{|\vec{r}_1 - \vec{r}_2|^3} \in \mathbb{Z} \quad (\text{Gauss 1833}).$$

A double integral that always returns an **integer**: it cannot change under continuous deformation — the **linking number**, the first invariant of knots and links.



[4.]

Von der *Geometria Situs*, die LEIBNITZ ahnte und in die nur einem Paar Geometern (EULER und VANDERMONDE) einen schwachen Blick zu thun vergönnt war, wissen und haben wir nach anderthalbhundert Jahren noch nicht viel mehr wie nichts.

Eine Hauptaufgabe aus dem *Grenzgebiet* der *Geometria Situs* und der *Geometria Magnitudinis* wird die sein, die Umschlingungen zweier geschlossener oder unendlicher Linien zu zählen.

Es seien die Coordinaten eines unbestimmten Punkts der ersten Linie x, y, z ; der zweiten x', y', z' und

$$\iint \frac{(x'-x)(dydz'-dzdy') + (y'-y)(dzdx'-dxdz') + (z-z')(dxdy'-dydx')}{[(x'-x)^2 + (y'-y)^2 + (z'-z)^2]^{\frac{3}{2}}} = V$$

dann ist dies Integral durch beide Linien ausgedehnt

$$= 4 m \pi$$

und m die Anzahl der Umschlingungen.

Der Werth ist gegenseitig, d. i. er bleibt derselbe, wenn beide Linien gegen einander umgetauscht werden. 1833. Jan. 22.



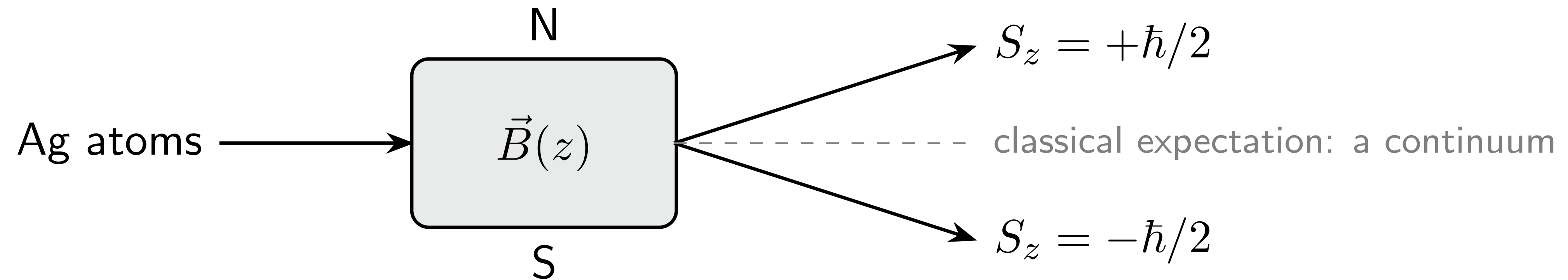
Gauss

SECTION 3

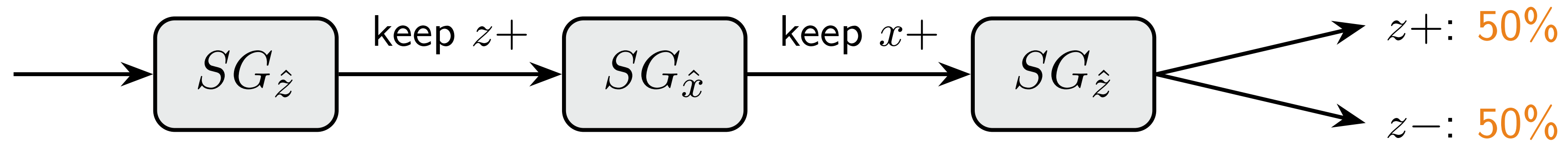
Quantum Mechanics

Stern-Gerlach experiment (1922)

Send silver atoms through a *non-uniform* magnetic field.



- **Classical prediction:** a continuous smear.
- **Observed:** exactly **two spins** (up and down) — the spin is quantized: $S_z = \pm\hbar/2$.



- Select atoms with $S_z = +\hbar/2$. Measure S_x : result $\pm\hbar/2$, 50/50.
- Keep $S_x = +\hbar/2$ and measure S_z *again*: 50/50 — the earlier z -information is gone!
- No classical bookkeeping (“the atom has definite S_z *and* S_x ”) can reproduce this.

What mathematical structure produces such behavior? **Linear algebra.**

Postulates for spin- $\frac{1}{2}$:

- A state is a unit vector in $\mathbb{C}^2$, e.g. $|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
- Spin observables are built from the Hermitian **Pauli matrices**: $S_i = \frac{\hbar}{2}\sigma_i$,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Their commutators close among themselves:

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k, \quad [S_i, S_j] = i\hbar \epsilon_{ijk} S_k.$$

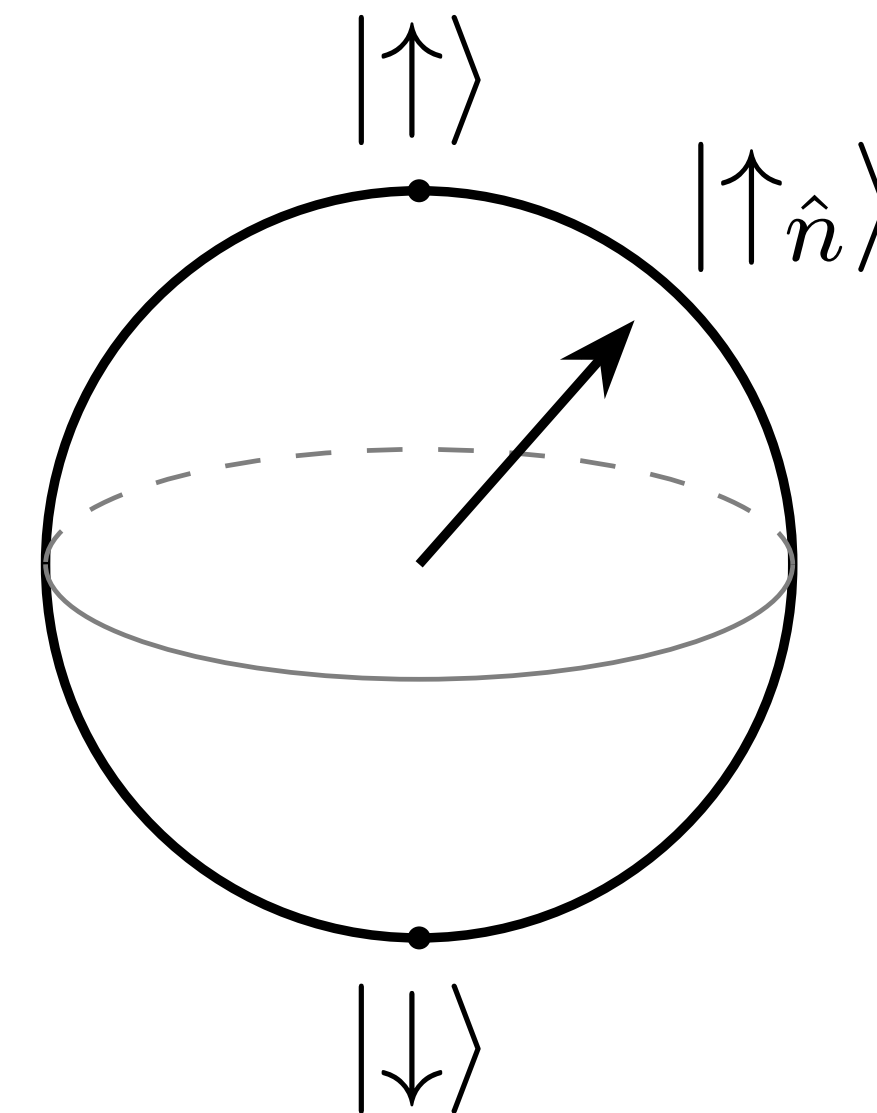
This relation ($\hbar = 1$) provides **su(2) Lie algebra**!

- Measurement outcomes = **eigenvalues** (each σ_i has eigenvalues $\pm 1 \Rightarrow S_i = \pm\hbar/2$: two spots!).

The spin along a unit vector $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ is $\hat{n} \cdot \vec{\sigma}$, with eigenvector

$$|\uparrow_{\hat{n}}\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

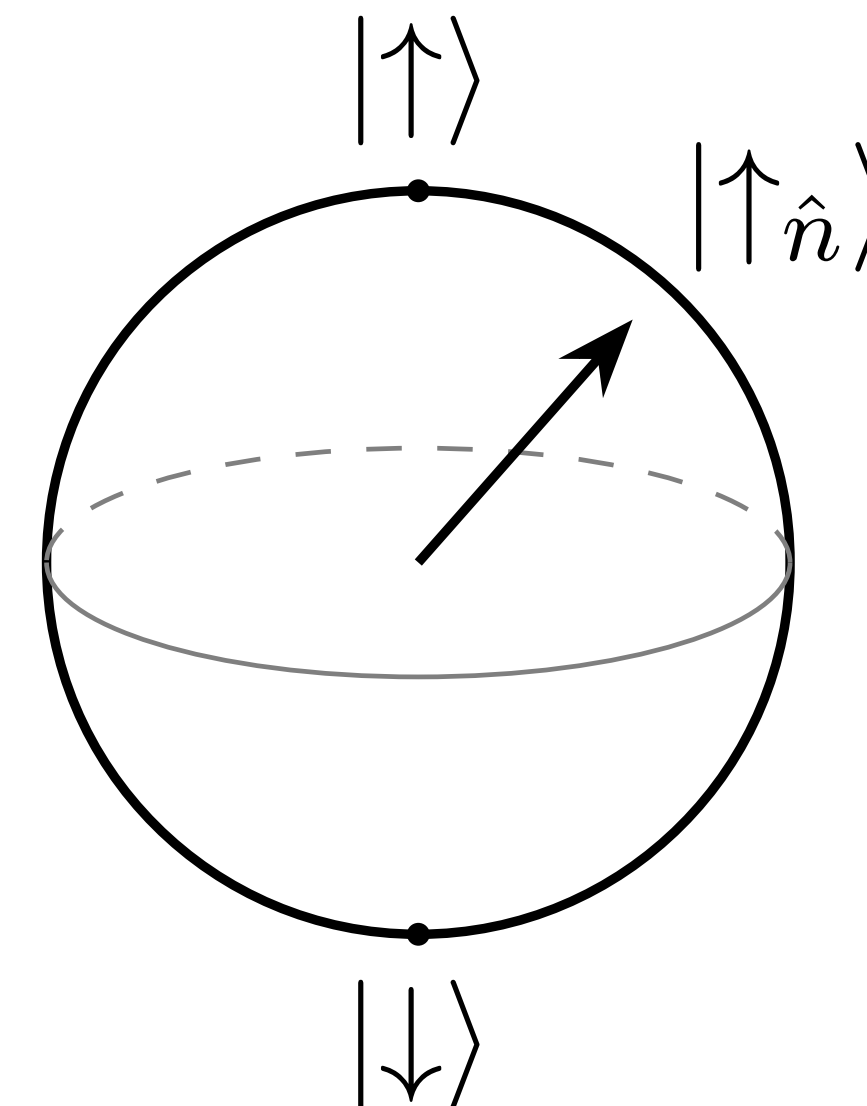
- Every spin state $\leftrightarrow$ a point of the sphere S^2 (the **Bloch sphere**).
- Today, such a two-level system is called a **qubit** — the unit of quantum information.



The spin along a unit vector $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ is $\hat{n} \cdot \vec{\sigma}$, with eigenvector

$$|\uparrow_{\hat{n}}\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

- A normalized spinor $(\alpha, \beta) \in \mathbb{C}^2$, $|\alpha|^2 + |\beta|^2 = 1$, is a point of S^3 , and the overall phase $e^{i\chi}$ changes no physics. The map $(\alpha, \beta) \mapsto \hat{n}$ is exactly the **Hopf map** $S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$ from Dirac's monopole: **the Bloch sphere is the base of the Hopf bundle.**



- Probability = |overlap with the eigenvector|²; afterwards, the state *becomes* that eigenvector.
- The **Pauli matrix** σ_x has eigenvectors $|\uparrow_x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $|\downarrow_x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
- Start with $|\uparrow\rangle$ (selected $z+$). Probability of $x+$: $|\langle\uparrow_x | \uparrow\rangle|^2 = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$ ✓
- The state is now $|\uparrow_x\rangle$. Probability of $z+$: $|\langle\uparrow | \uparrow_x\rangle|^2 = \frac{1}{2}$ ✓ — the memory of $z+$ is erased.
- **Non-commuting operators**

$$[S_x, S_z] = -i\hbar S_y.$$

imply incompatible measurements. This is the seed of the uncertainty principle.

Uncertainty from non-commuting operators

- A quantum observable is a Hermitian matrix (operator). Its **uncertainty** in the state $|\psi\rangle$ is its spread:

$$(\Delta A)^2 = \langle A^2 \rangle_\psi - \langle A \rangle_\psi^2.$$

- If A and B were both perfectly sharp in the same state, then

$$A|\psi\rangle = a|\psi\rangle, \quad B|\psi\rangle = b|\psi\rangle \quad \implies \quad [A, B]|\psi\rangle = 0.$$

Therefore, when $[A, B]|\psi\rangle \neq 0$, the two observables *cannot both be sharp*.

- Quantitatively, the **Robertson uncertainty relation** is

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle_\psi|.$$

- For spin, $[S_x, S_y] = i\hbar S_z$, so

$$\Delta S_x \Delta S_y \geq \frac{\hbar}{2} |\langle S_z \rangle|.$$

Position–momentum uncertainty

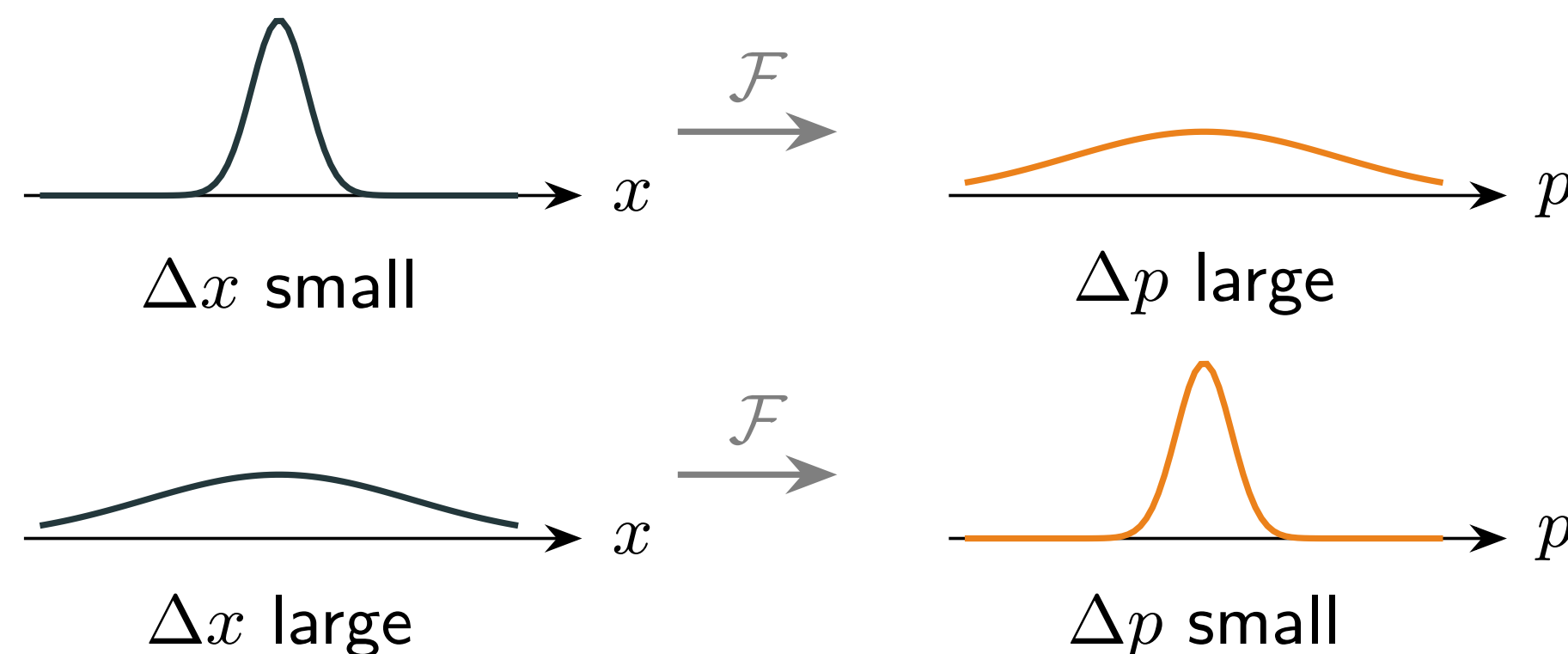
- Position and momentum *do not commute*: $[\hat{x}, \hat{p}] = i\hbar$. The Robertson relation then gives the **uncertainty principle**

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad \text{— they cannot both be sharp.}$$

- The reason is Fourier: the momentum wave function is the **Fourier transform** of $\psi(x)$ — the same transformation that diagonalized our springs:

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \psi(x) e^{-ipx/\hbar} dx,$$

and a function and its Fourier transform *cannot both be narrow*:



The Schrödinger equation and time–energy uncertainty

- In the position representation the momentum operator becomes $\hat{p} = -i\hbar \frac{d}{dx}$ — after a Fourier transformation, $-i\hbar d/dx$ is simply multiplication by p .
- The energy then becomes the **Hamiltonian operator**

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x),$$

and the dynamics of ψ is the **Schrödinger equation** $i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$.

- **Time–energy:** time is a *parameter*, not an operator. From the Schrödinger equation one derives (Mandelstam–Tamm)

$$\Delta H \tau_A \geq \frac{\hbar}{2}, \quad \tau_A = \frac{\Delta A}{|d\langle A \rangle / dt|} = \text{time for } \langle A \rangle \text{ to shift by one } \Delta A.$$

A state with a small energy spread can only change *slowly*.

Modern Quantum Mechanics

J. J. Sakurai



Revised Edition

J. J. サクライ著, ナポリターノ編・著
第 3 版
現代の量子力学(上)

桜井明夫 共訳
常次宏 一

物理学叢書
112

吉岡書店

SECTION 4

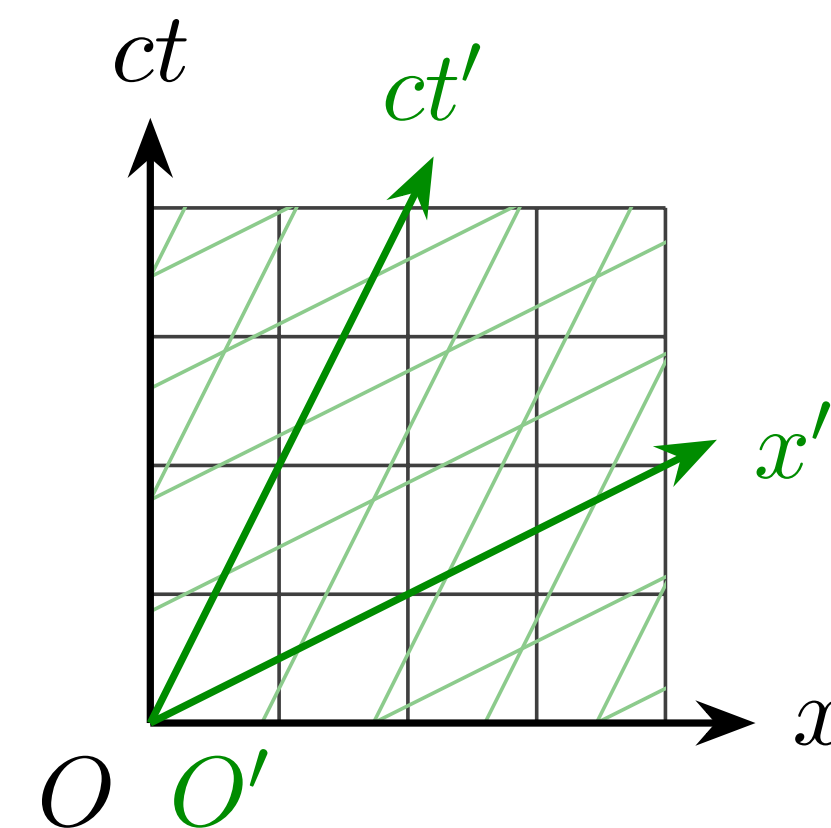
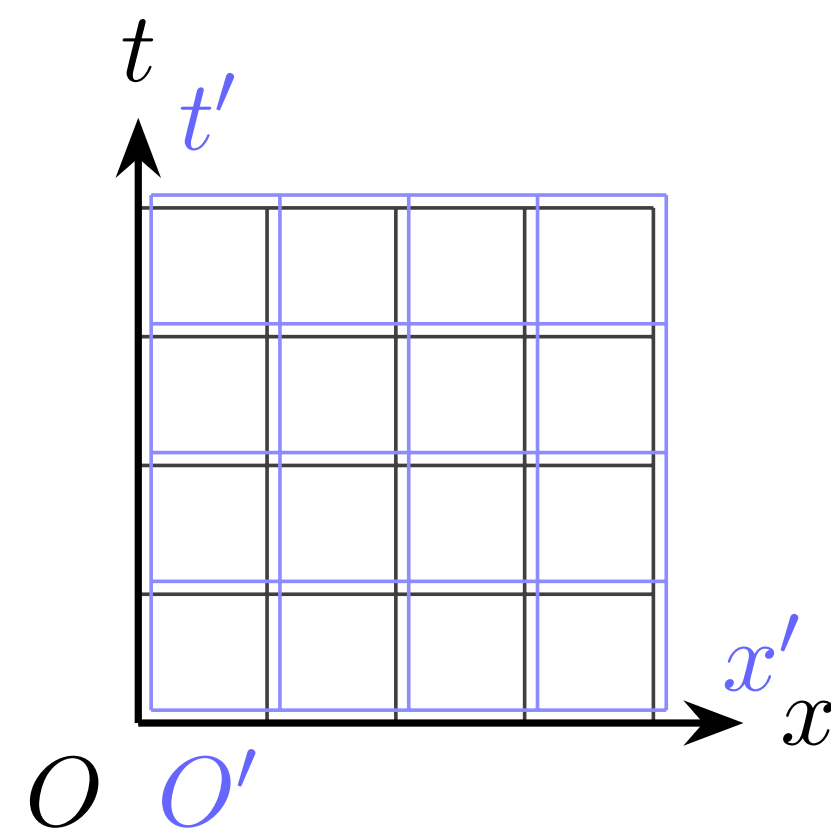
Special Relativity

From the Galilei transformation to the Lorentz transformation

Einstein's postulates (1905)

1. The laws of physics are the same in every inertial frame.
 2. The speed of light c is the same for *all* inertial observers.
- **Galilei:** time and distances are *separately* invariant: $t' = t$, $x'_2 - x'_1 = x_2 - x_1$ — the two grids coincide.
 - **Lorentz:** light-speed invariance forces a different invariant: only the *mixture*

$$(x'_2 - x'_1)^2 - c^2(t'_2 - t'_1)^2 = (x_2 - x_1)^2 - c^2(t_2 - t_1)^2$$

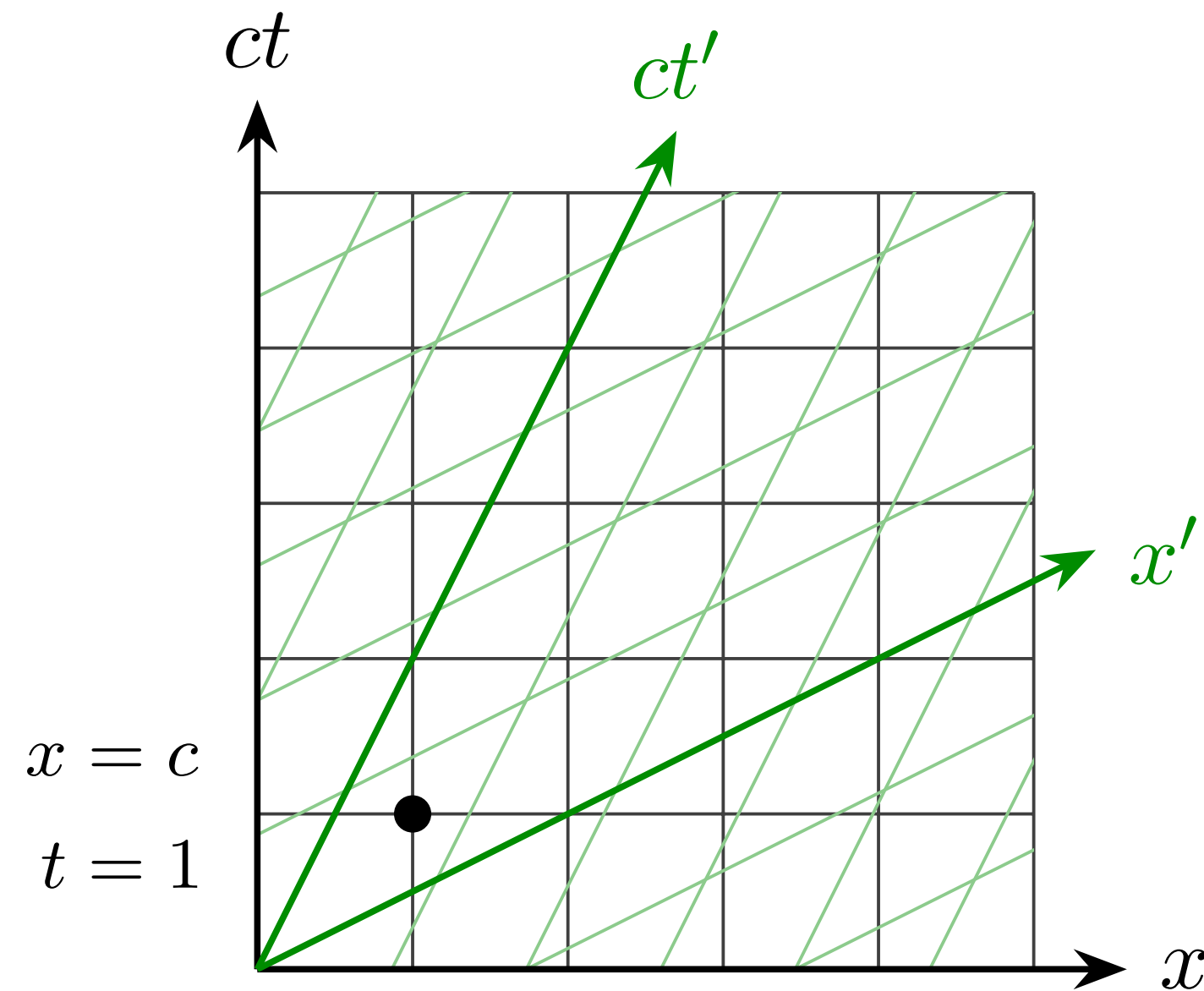


- Both transformations are *linear* — 2×2 matrices:

$$\underbrace{\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}}_{\text{rotation: preserves } x^2 + y^2}$$

$$\underbrace{\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \cosh \eta & -\sinh \eta \\ -\sinh \eta & \cosh \eta \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}}_{\text{boost: preserves } c^2 t^2 - x^2}$$

- A boost is a *hyperbolic rotation* of spacetime; the “angle” η is fixed by the velocity: $\tanh \eta = v/c$ and $\cosh \eta = 1/\sqrt{1 - (v/c)^2}$.
- Example:



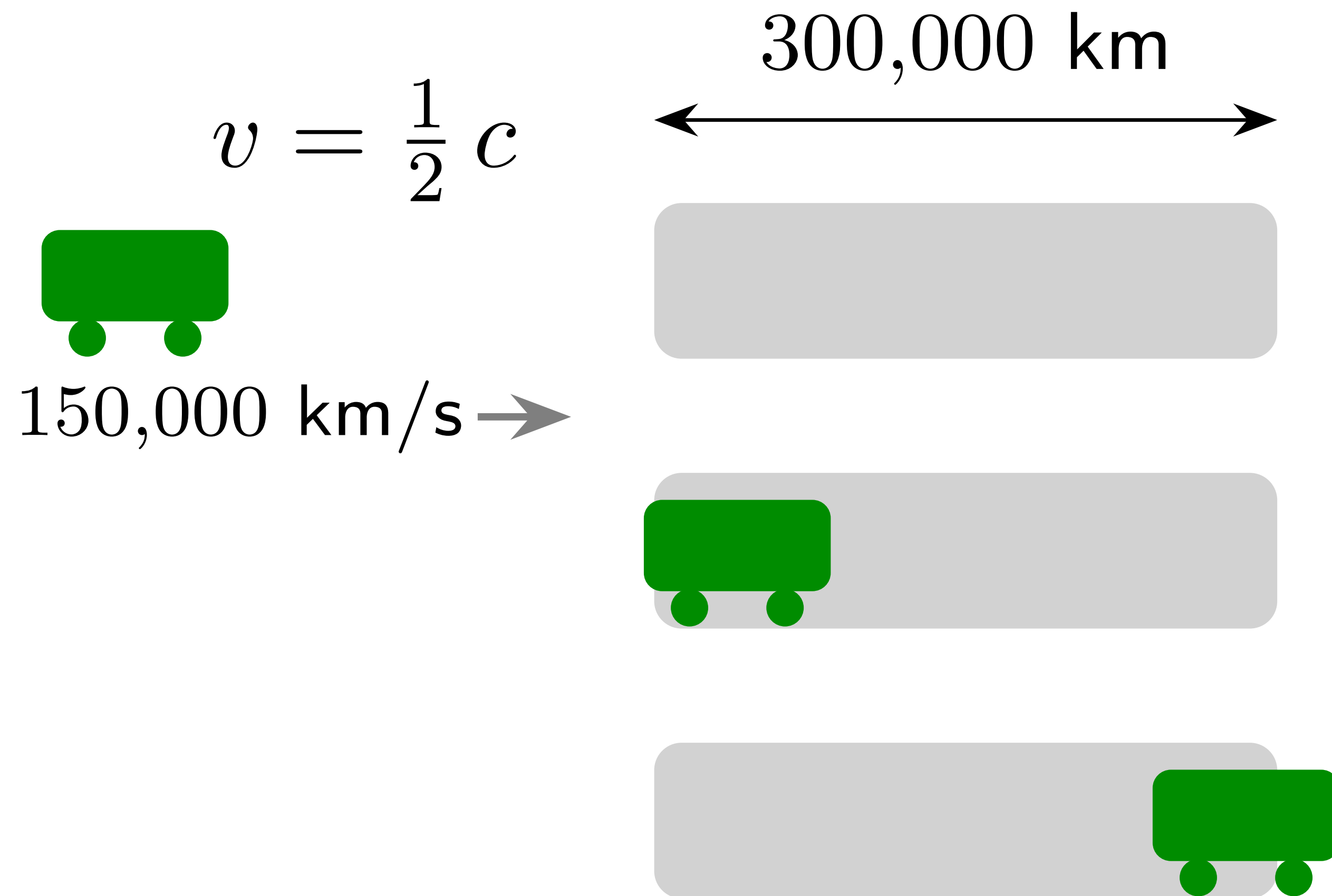
Inertial frame S

Inertial frame S'

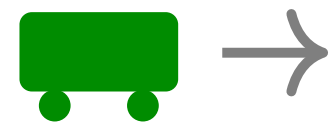
$$v = \frac{1}{2} c$$

$$x' = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} (x - vt) = \frac{1}{\sqrt{3}} c$$

$$t' = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \left(-\frac{v}{c^2} x + t \right) = \frac{1}{\sqrt{3}}$$



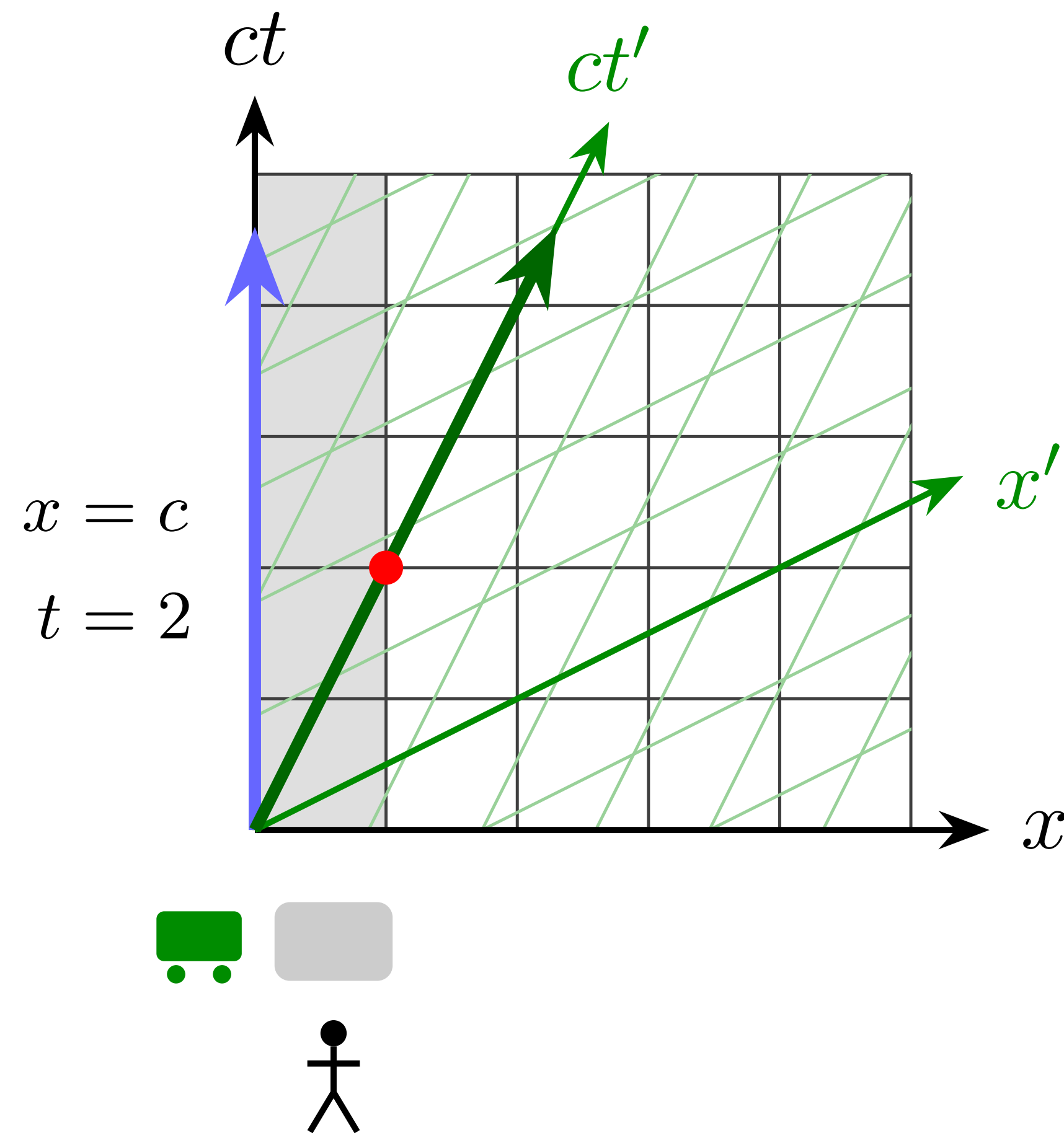
150,000 km/s



300,000 km



Inertial frame S



Inertial frame S'

$$t' = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \left(-\frac{v}{c^2} x + t \right)$$

$$= 1.73$$

The train passes through the tunnel in 1.73 s.

150,000 km/s



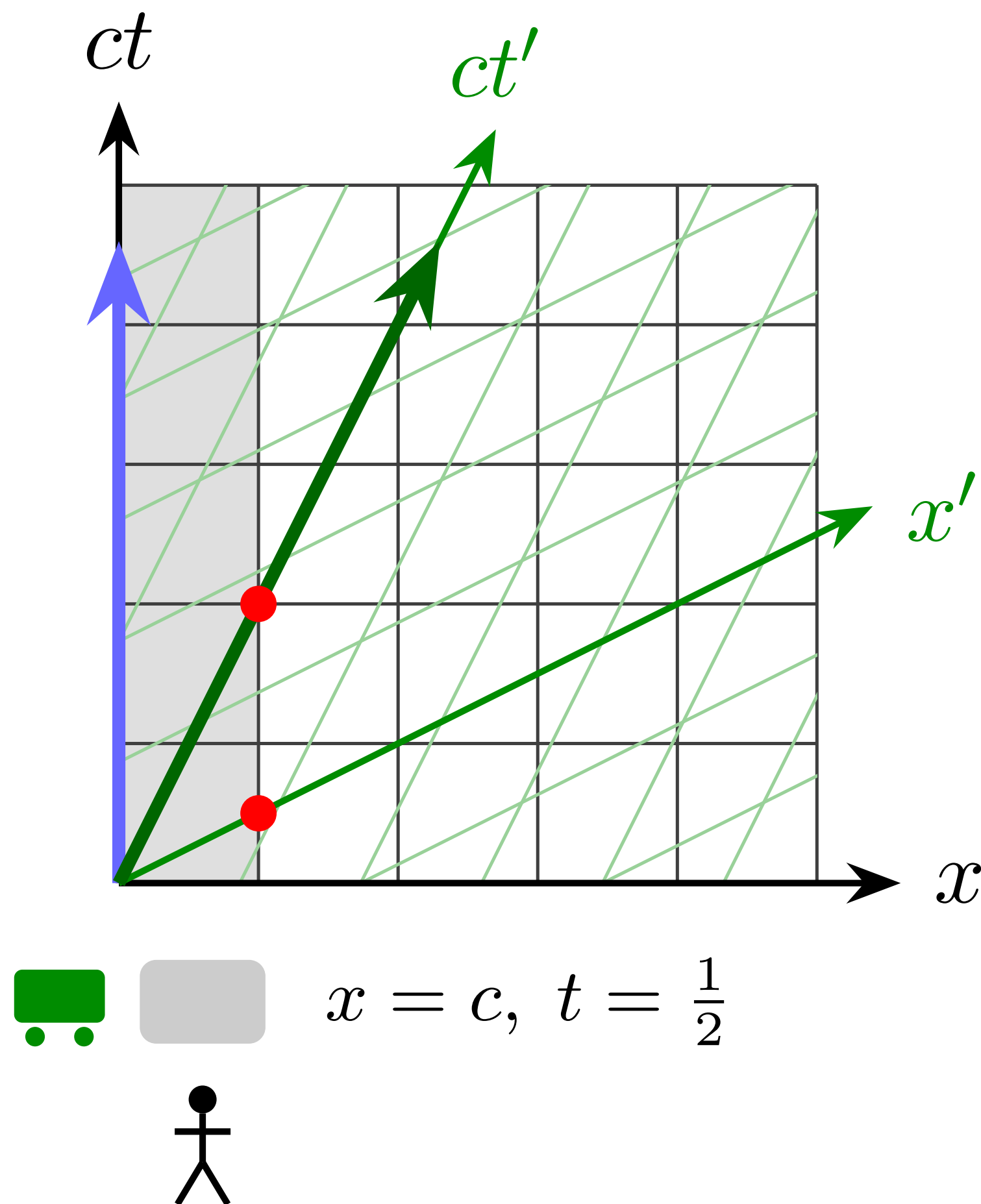
300,000 km



Inertial frame S

$$x = c, t = 2$$

$$t' = 1.73$$



Inertial frame S'

$$x' = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} (x - vt)$$

$$= \frac{\sqrt{3}}{2} c = 260,000 \text{ km}$$

For the train, the tunnel is 260,000 km long.

Boosts and rotations were *real* 2×2 matrices — the *complex* 2×2 matrices hold the full story.

- Package a spacetime point into a Hermitian 2×2 matrix (the Pauli matrices again!):

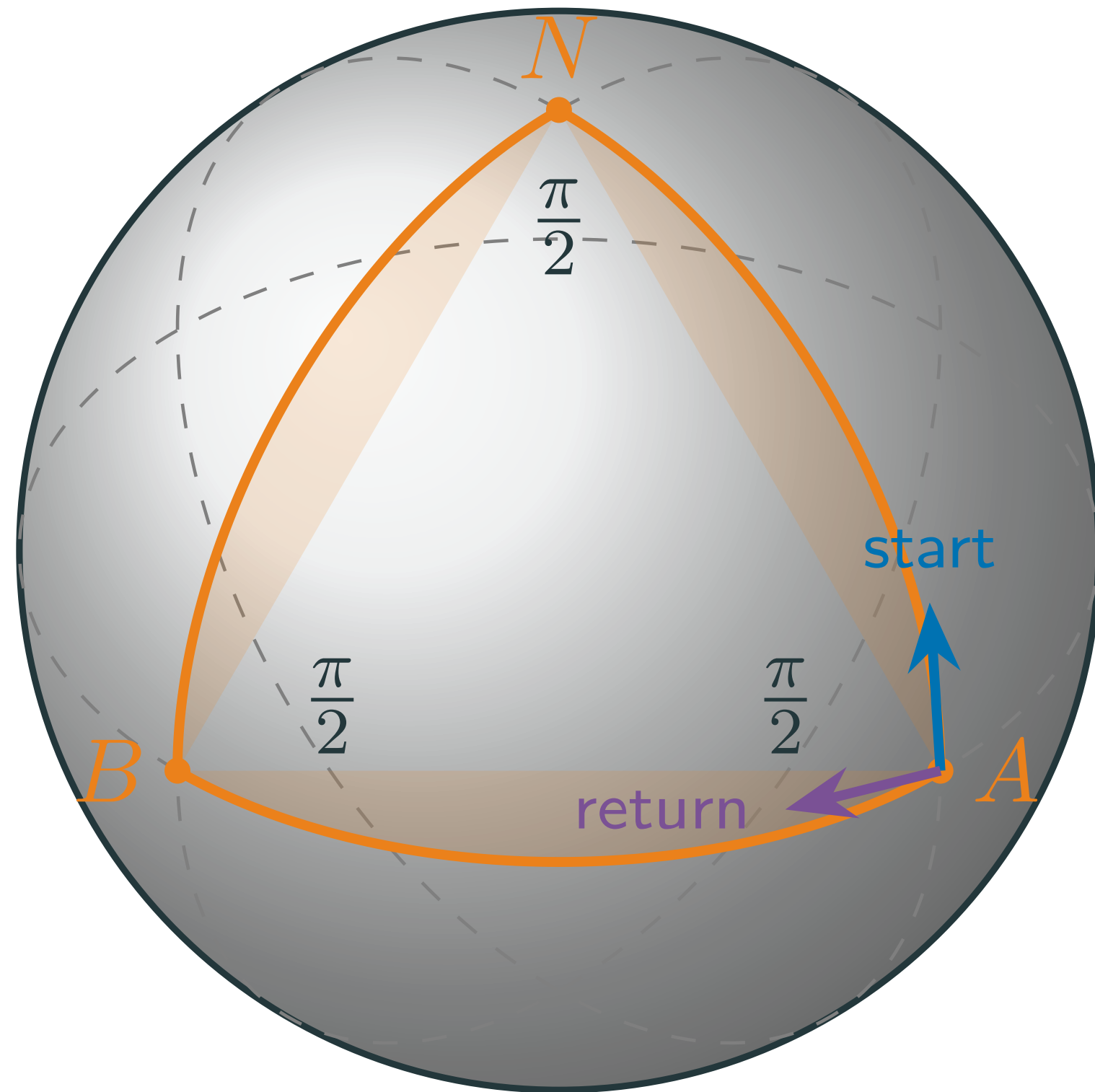
$$X = ct \mathbf{1} + \vec{x} \cdot \vec{\sigma} = \begin{pmatrix} ct + z & x - iy \\ x + iy & ct - z \end{pmatrix}, \quad \det X = c^2 t^2 - x^2 - y^2 - z^2.$$

- For any $M \in SL(2, \mathbb{C})$ (complex 2×2 , $\det M = 1$), the map $X \mapsto MXM^\dagger$ is linear, keeps X Hermitian, and preserves $\det X \Rightarrow$ it is a Lorentz transformation.
- In fact *every* Lorentz transformation (keeping the directions of time and space) arises this way, with M and $-M$ giving the same one: $SL(2, \mathbb{C}) / \{\pm 1\} \cong L_+^\uparrow$.

SECTION 5

General Relativity

Curved space



The round sphere of unit radius has

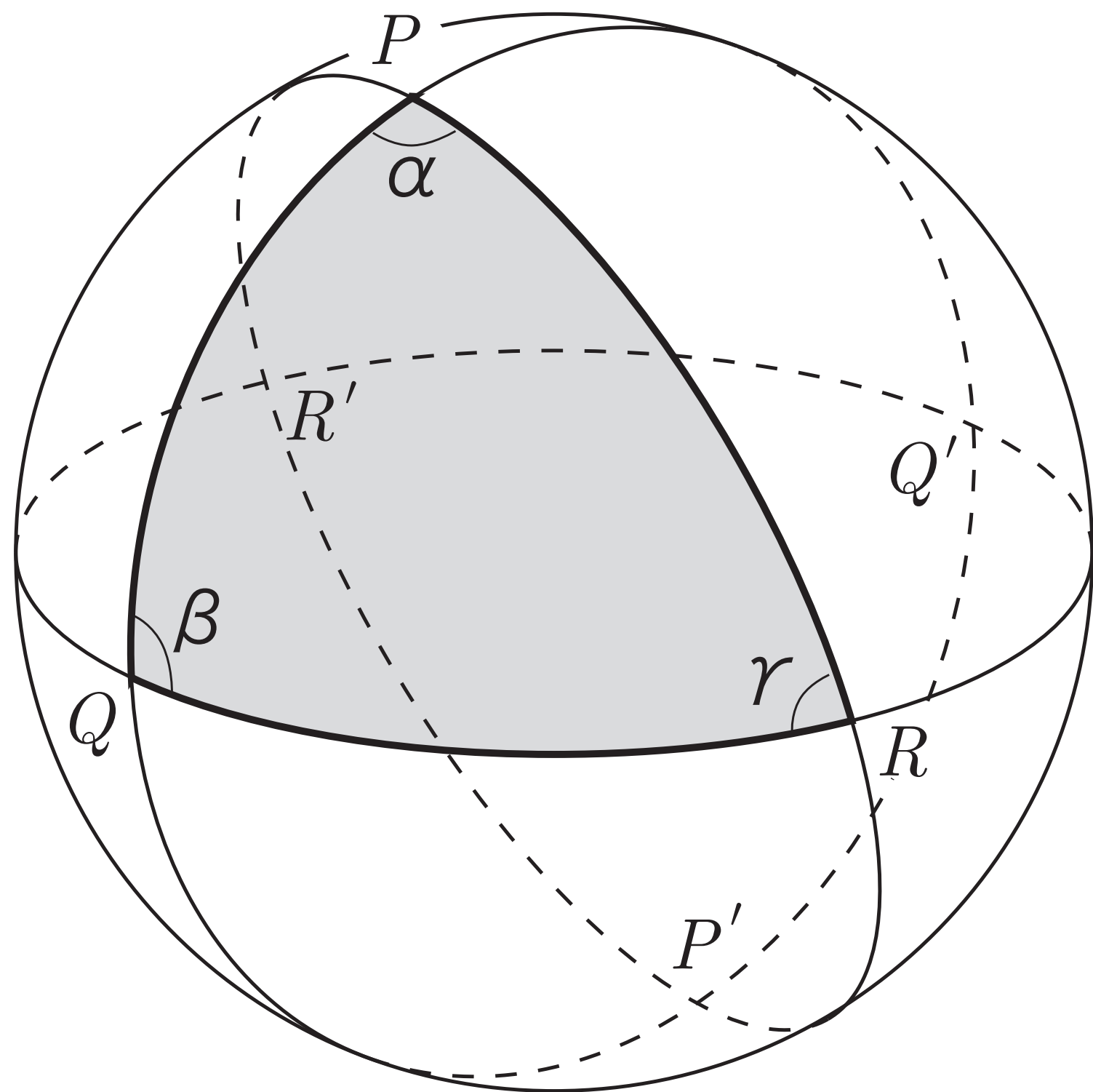
$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2, \quad K = 1.$$

Three orthogonal coordinate planes cut out an octant whose sides are **great-circle geodesics**.

$$\text{Area}(\triangle) = \frac{\pi}{2}.$$

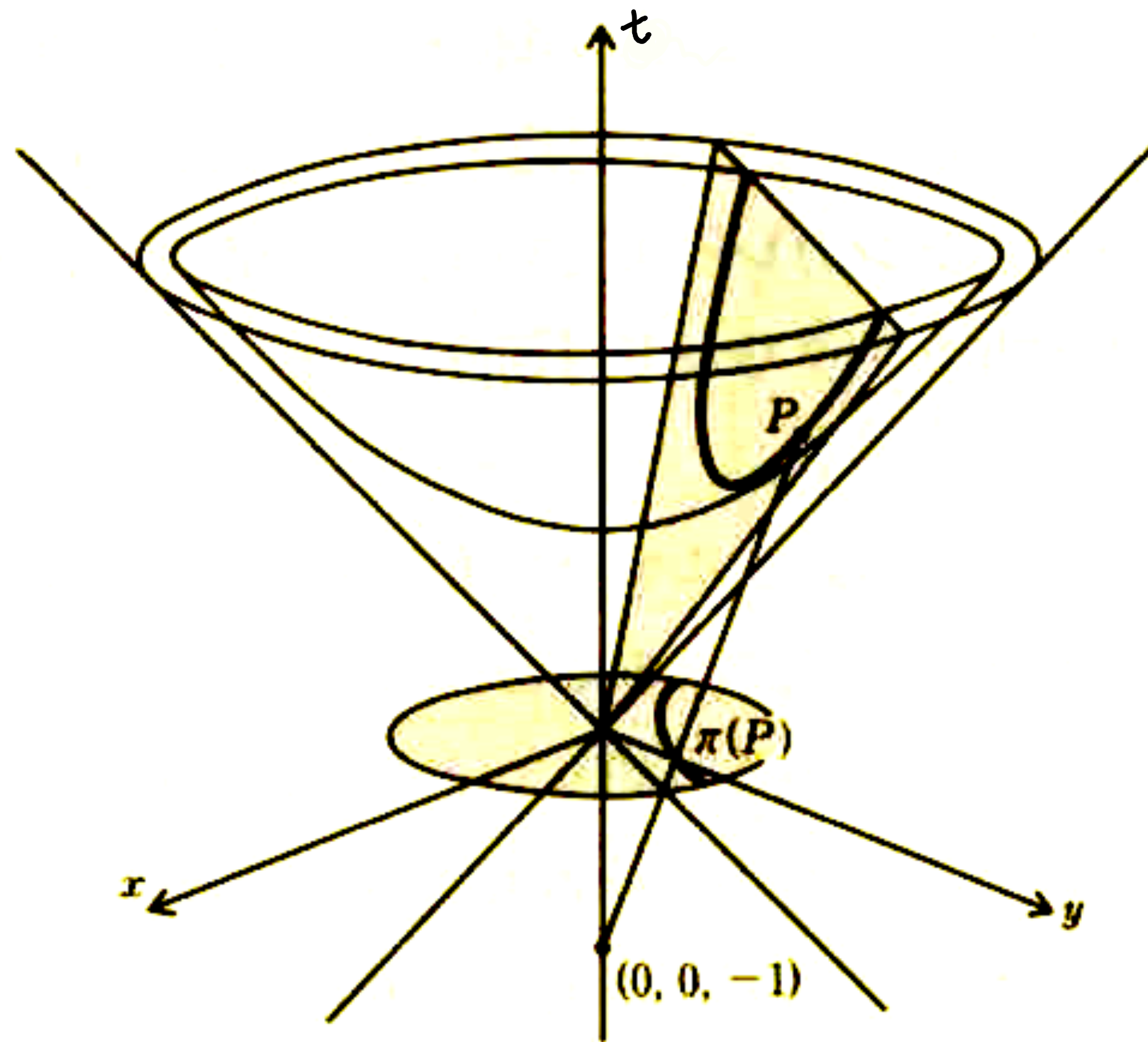
Parallel transport around $A \rightarrow B \rightarrow N \rightarrow A$ returns a vector rotated by

$$\Delta\psi = \int_{\triangle} K dA = \text{Area}(\triangle) = \frac{\pi}{2}.$$



$$\text{Area}(\triangle) = \alpha + \beta + \gamma - \pi$$

Hyperboloid and Poincare disk



- In 3D Minkowski space, the surface

$$t^2 - x^2 - y^2 = 1, \quad (t = \cosh \rho, \quad x + iy = \sinh \rho e^{i\phi})$$

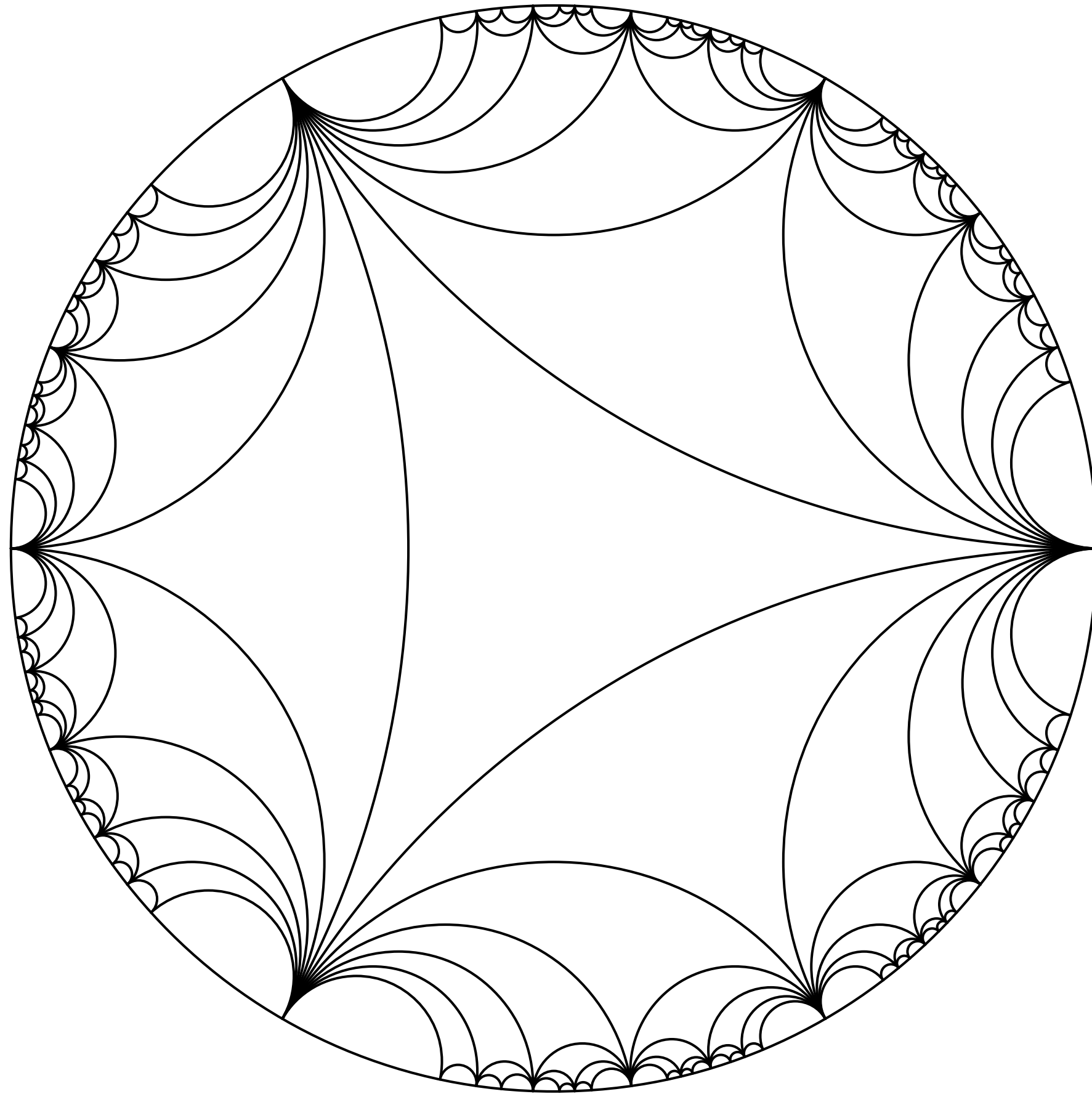
inherits the metric $d\ell^2 = d\rho^2 + \sinh^2 \rho d\phi^2$,
of constant curvature $K = -1$.

- Projecting from $(X^0, x, y) = (-1, 0, 0)$ maps the sheet onto the unit disk $|z| < 1$ with $|z| = \tanh(\rho/2)$, giving the **Poincaré metric**

$$d\ell^2 = \frac{4 |dz|^2}{(1 - |z|^2)^2}.$$

- **Geodesics** are diameters and circular arcs *orthogonal* to the boundary.

Hyperboloid and Poincare disk



- In 3D Minkowski space, the surface

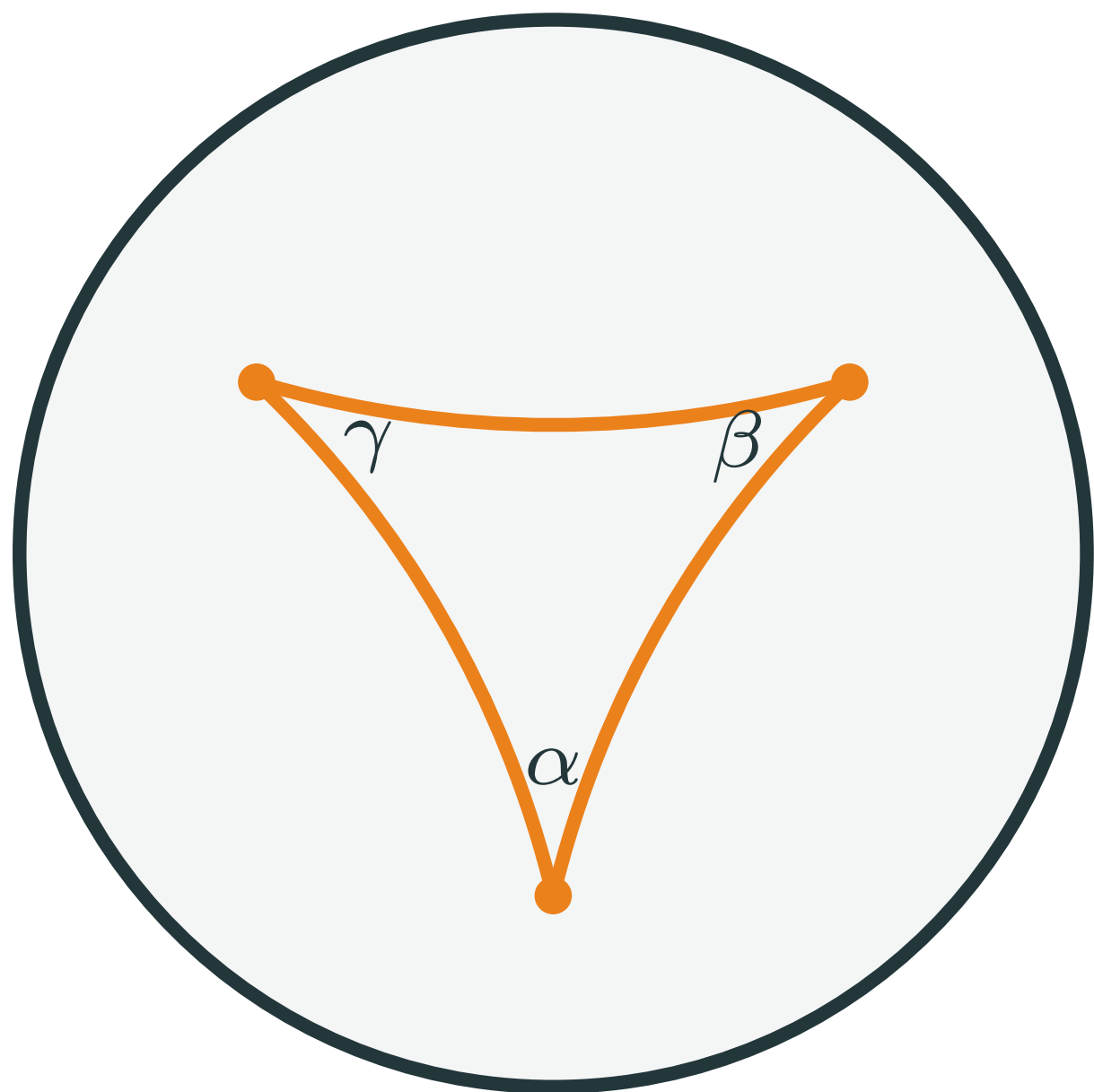
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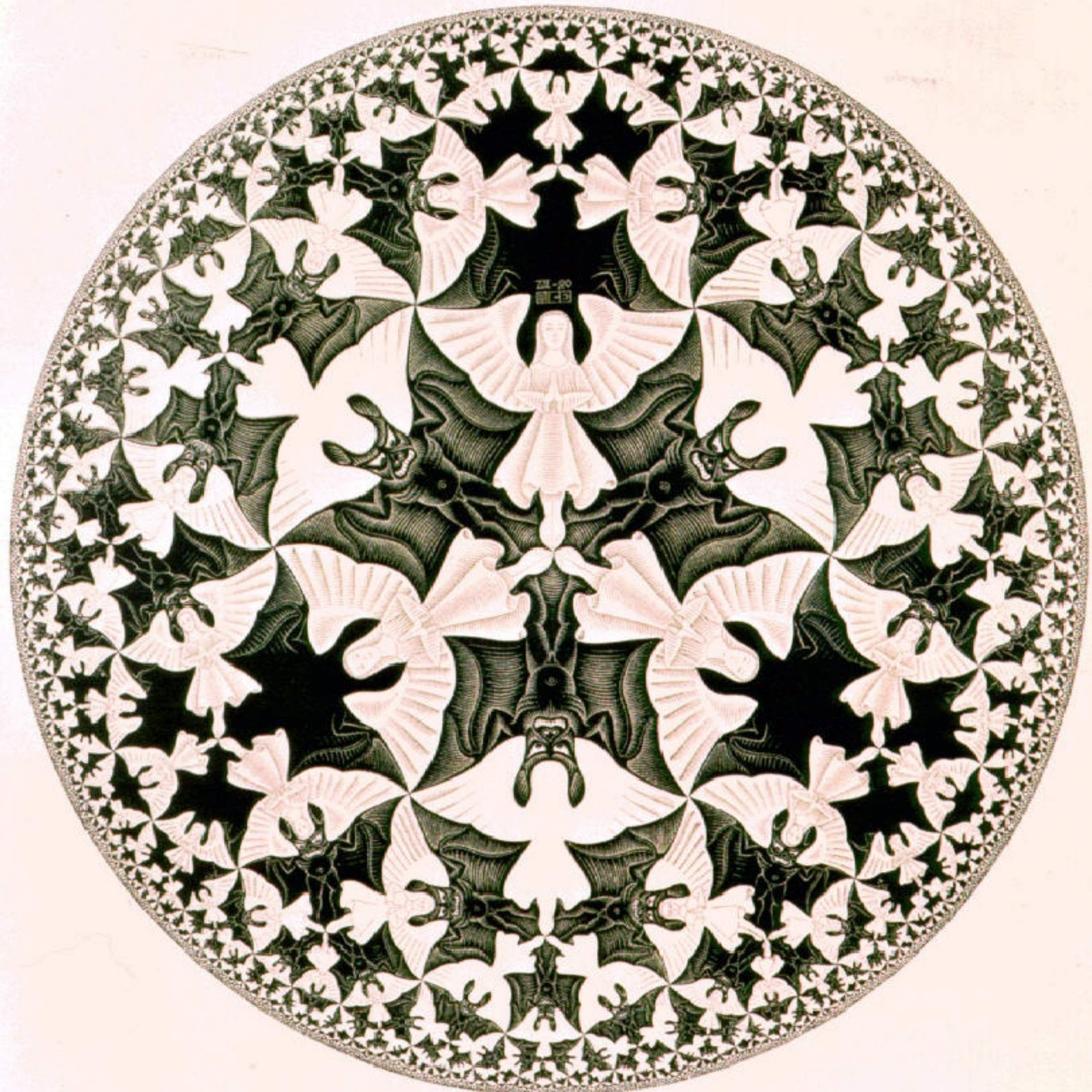
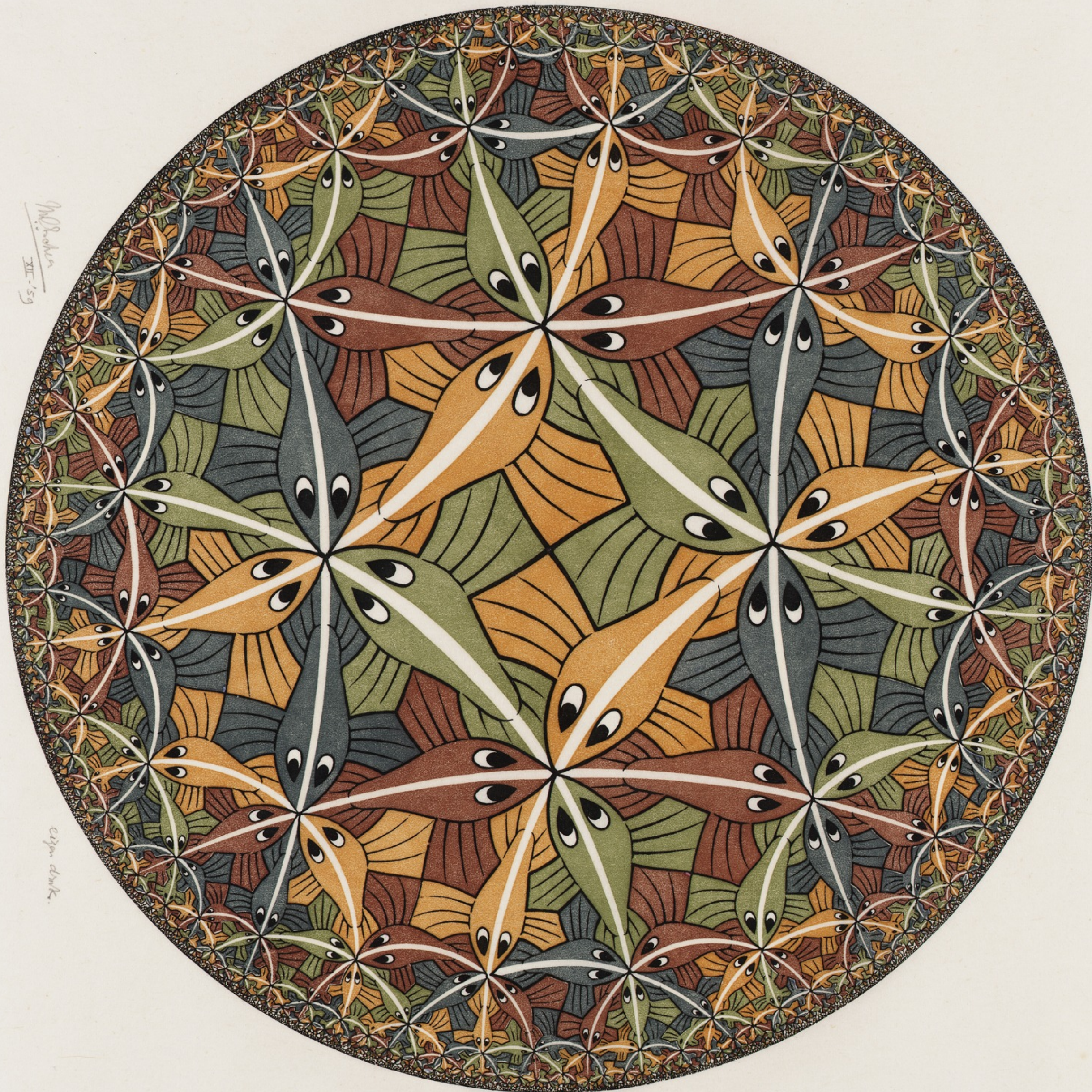
$$d\ell^2 = \frac{4 |dz|^2}{(1 - |z|^2)^2}.$$

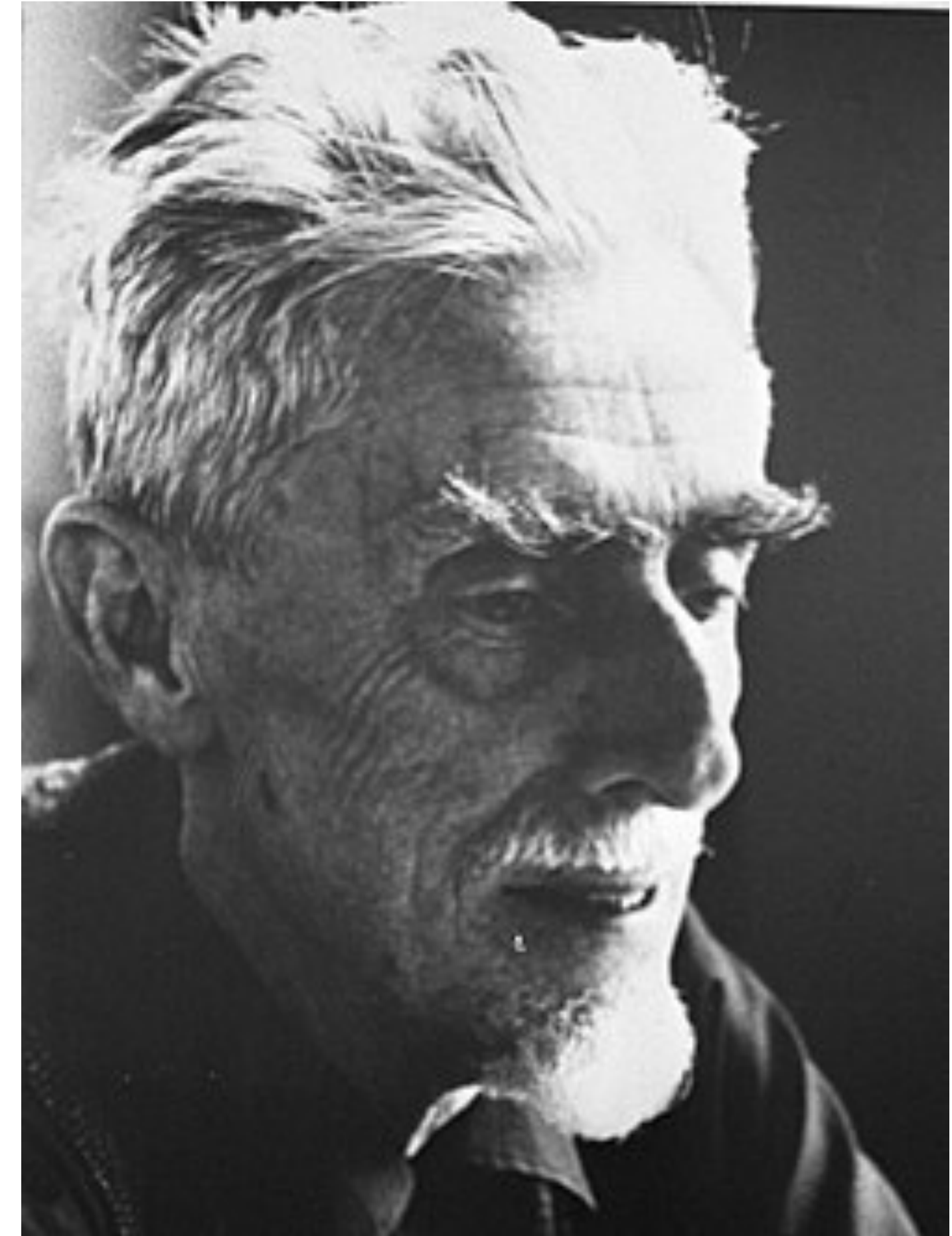
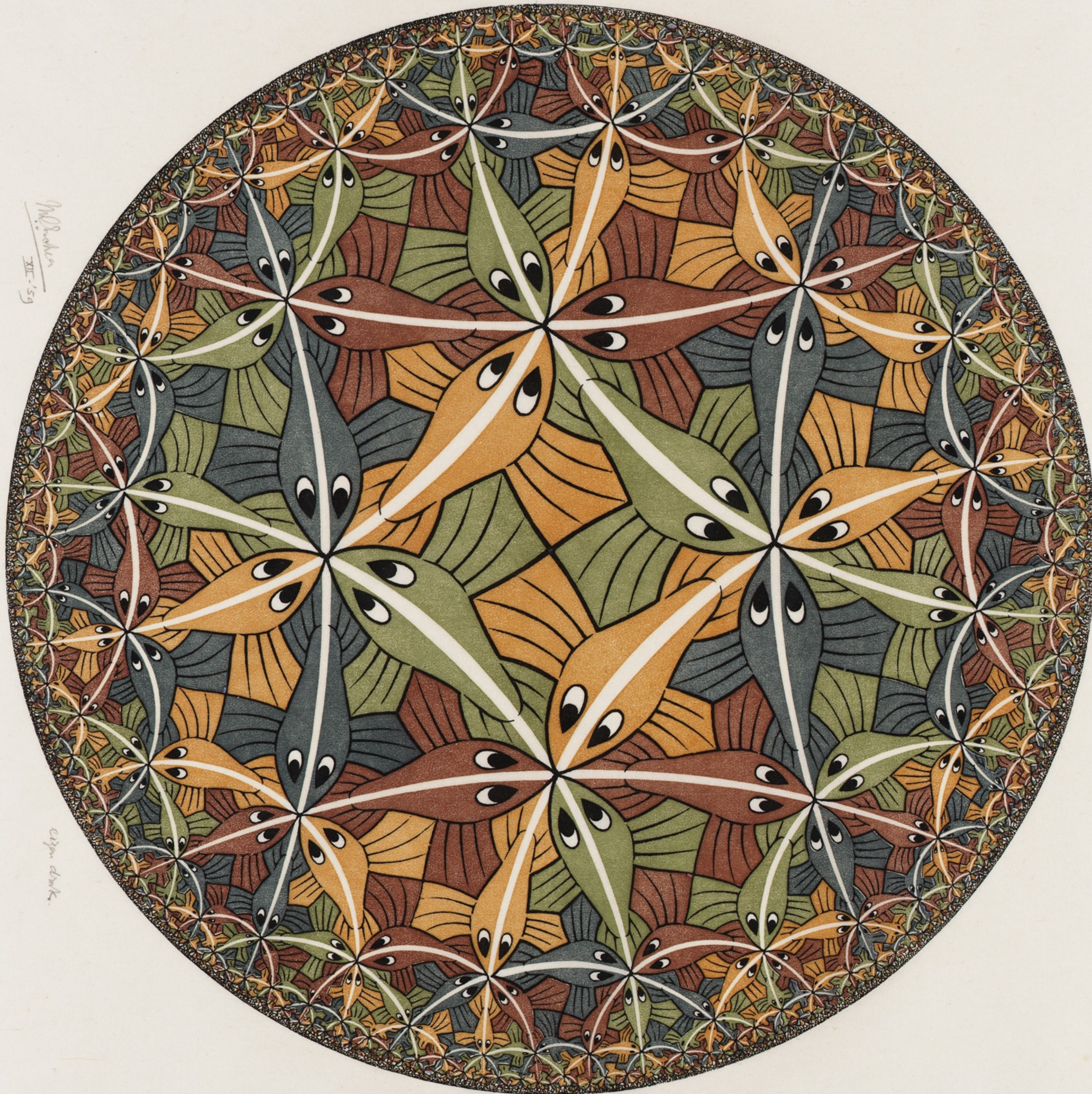
- **Geodesics** are diameters and circular arcs *orthogonal* to the boundary.



Poincaré disk

$$\text{Area}(\triangle) = \pi - (\alpha + \beta + \gamma)$$





M.C. Escher

Riemannian geometry: one field $g_{\mu\nu}$ rules everything

- A **Riemannian manifold** is a space that locally looks like $\mathbb{R}^n$, with a **metric** — a position-dependent ruler:

$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu \quad (\text{sphere: } g_{\theta\theta} = R^2, g_{\phi\phi} = R^2 \sin^2 \theta).$$

- **Geodesics** — the straightest paths, extremizing $\int ds$:

$$\frac{d^2 x^\rho}{d\tau^2} + \Gamma^\rho_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0, \quad \Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} (\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\mu\lambda} - \partial_\lambda g_{\mu\nu}),$$

with the **Christoffel symbols** Γ built from the metric alone.

- **Riemann curvature** — how parallel transport around a small loop rotates a vector ($\Delta\psi$ on the sphere; on a surface it is Gauss's K , and flat $\Leftrightarrow R^\rho_{\sigma\mu\nu} = 0$):

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}, \quad R_{\mu\nu} = R^\rho_{\mu\rho\nu}, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

General covariance + equivalence principle $\Rightarrow$ Einstein's equation

equivalence principle

in free fall gravity vanishes: $g_{\mu\nu}(p) = \eta_{\mu\nu}$, $\Gamma^\rho{}_{\mu\nu}(p) = 0$
only tidal gravity ($\partial\Gamma \sim$ **curvature**) survives

general covariance

laws take the same form in every coordinate system
 $\Rightarrow$ they must be **tensor** equations

gravity is geometry: a tensor equation for $g_{\mu\nu}$, sourced by energy–momentum $T_{\mu\nu}$

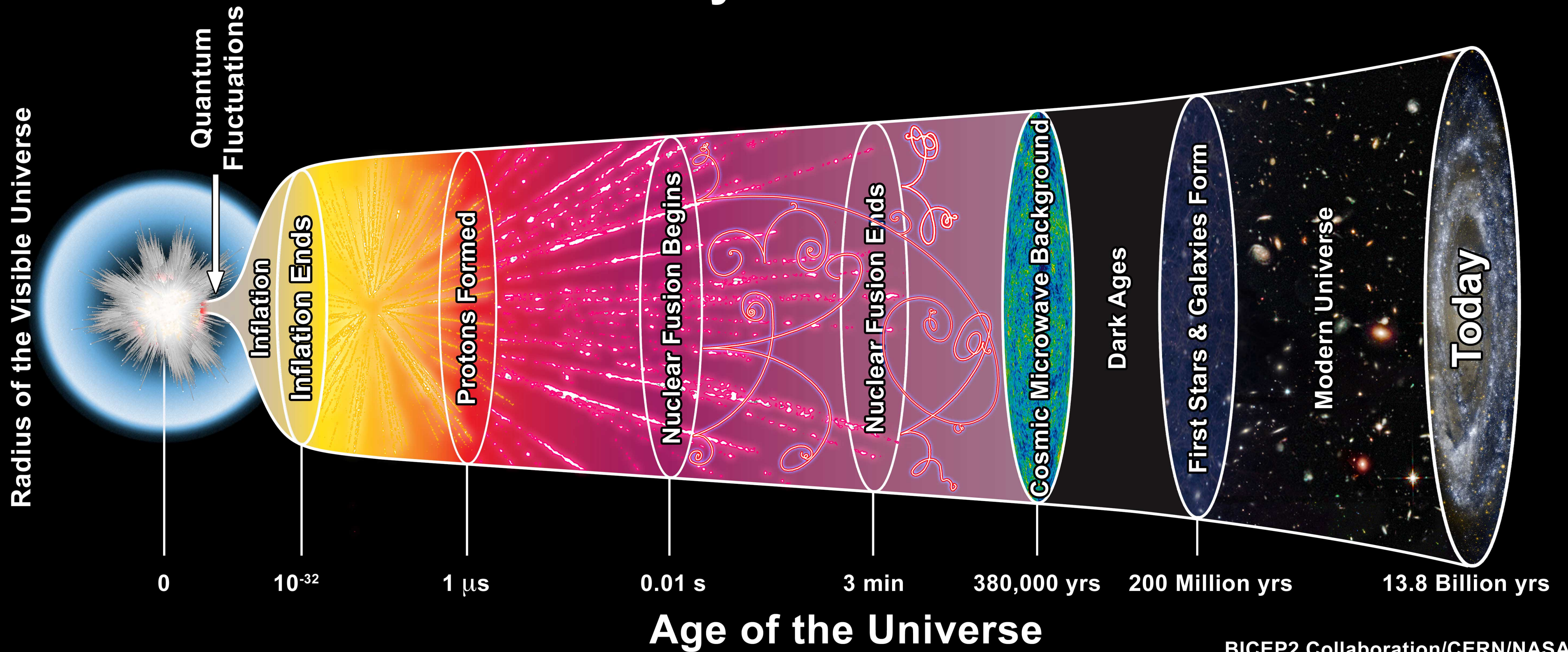
Conservation $\nabla^\mu T_{\mu\nu} = 0$ demands a *divergence-free* tensor built from $g_{\mu\nu}$ with at most two derivatives — and the Bianchi identity provides exactly one (uniquely so: Lovelock 1971):

$$\nabla^\mu \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0.$$

Recovering Newton's $\nabla^2\Phi = 4\pi G\rho$ for weak fields fixes the constant:

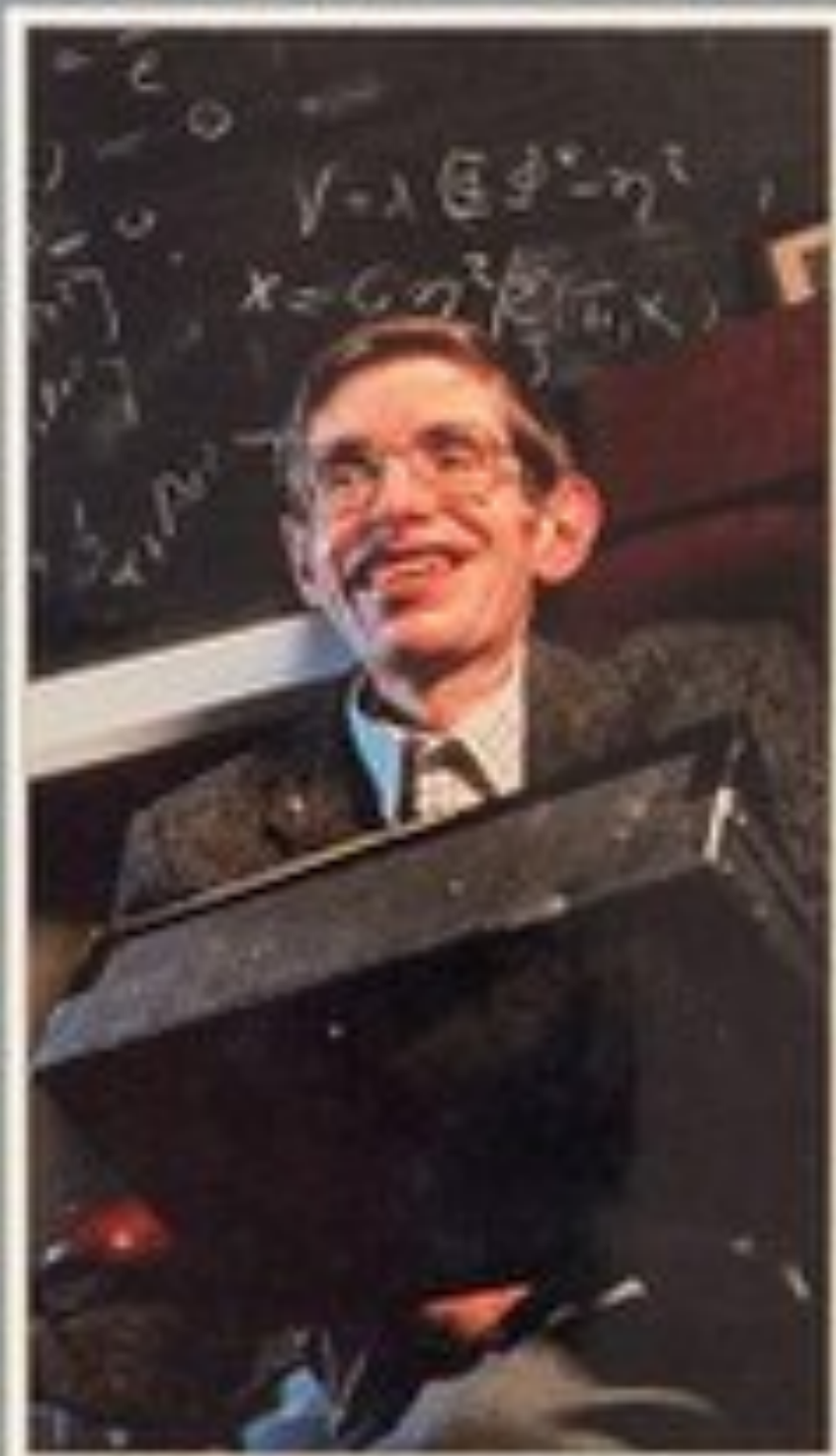
$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (\text{Einstein 1915}).$$

History of the Universe



A BRIEF HISTORY OF TIME

FROM
THE BIG
BANG TO
BLACK
HOLES



**STEPHEN
W. HAWKING**

INTRODUCTION BY CARL SAGAN

THE FIRST THREE MINUTES

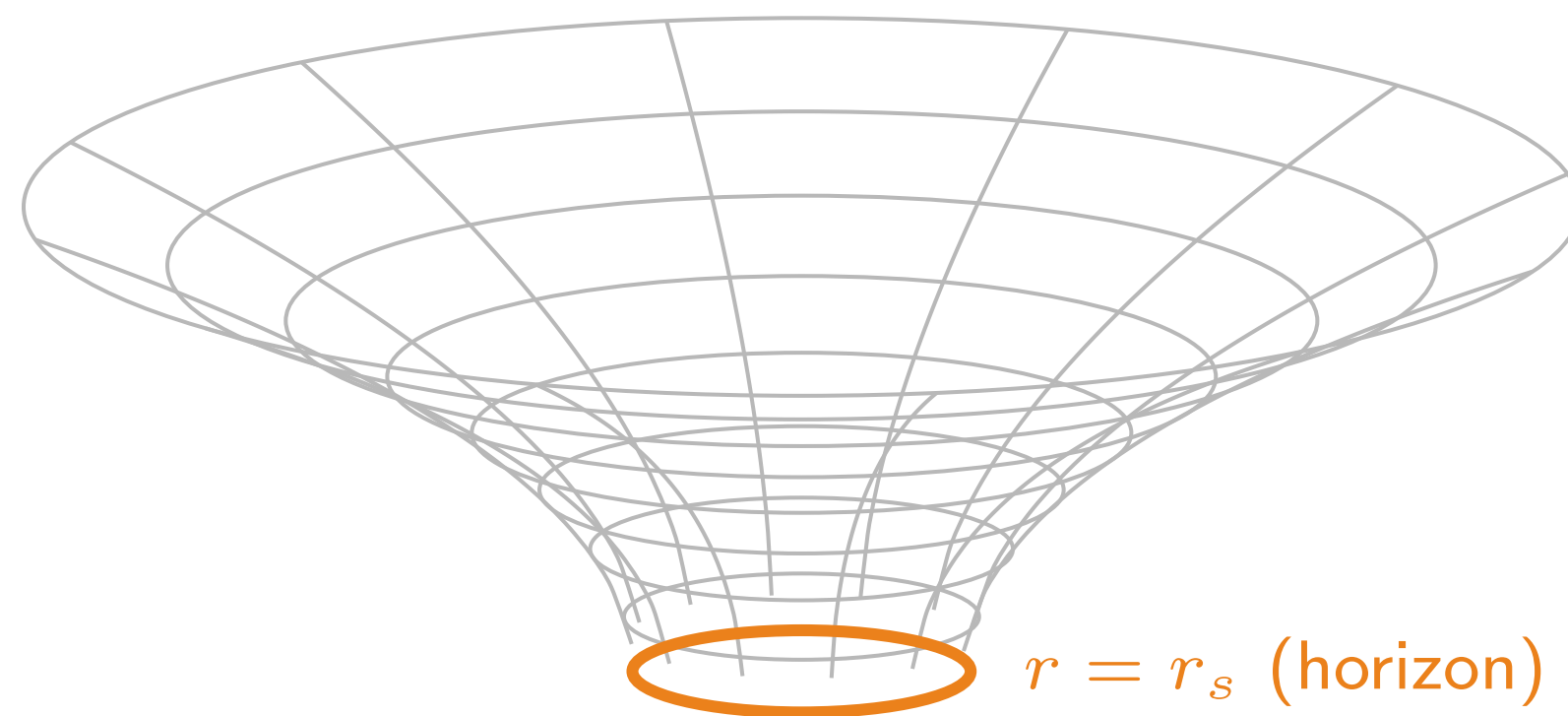
STEVEN WEINBERG

A modern view of the origin
of the universe

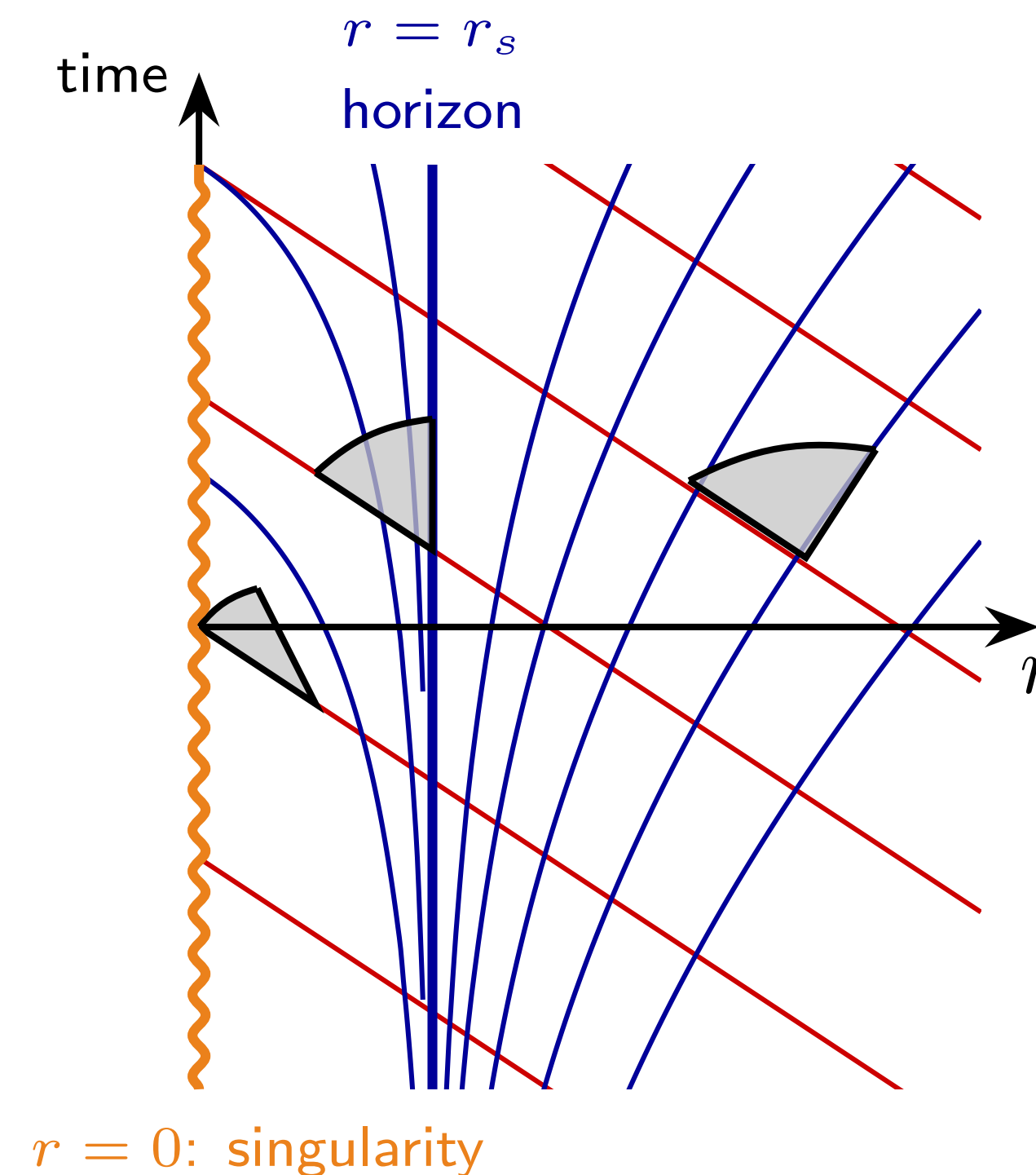
The Schwarzschild black hole and its causal structure

The vacuum solution outside a spherical mass M (Schwarzschild 1916), with $r_s = 2GM/c^2$:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2).$$

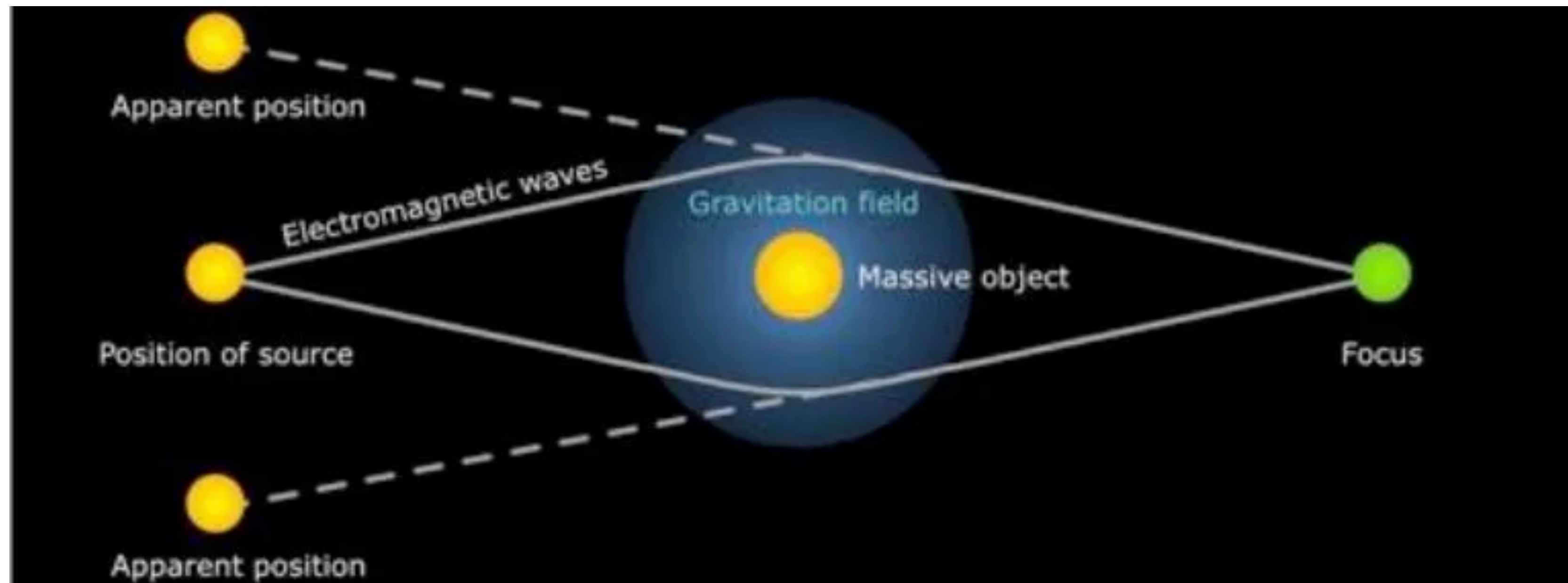


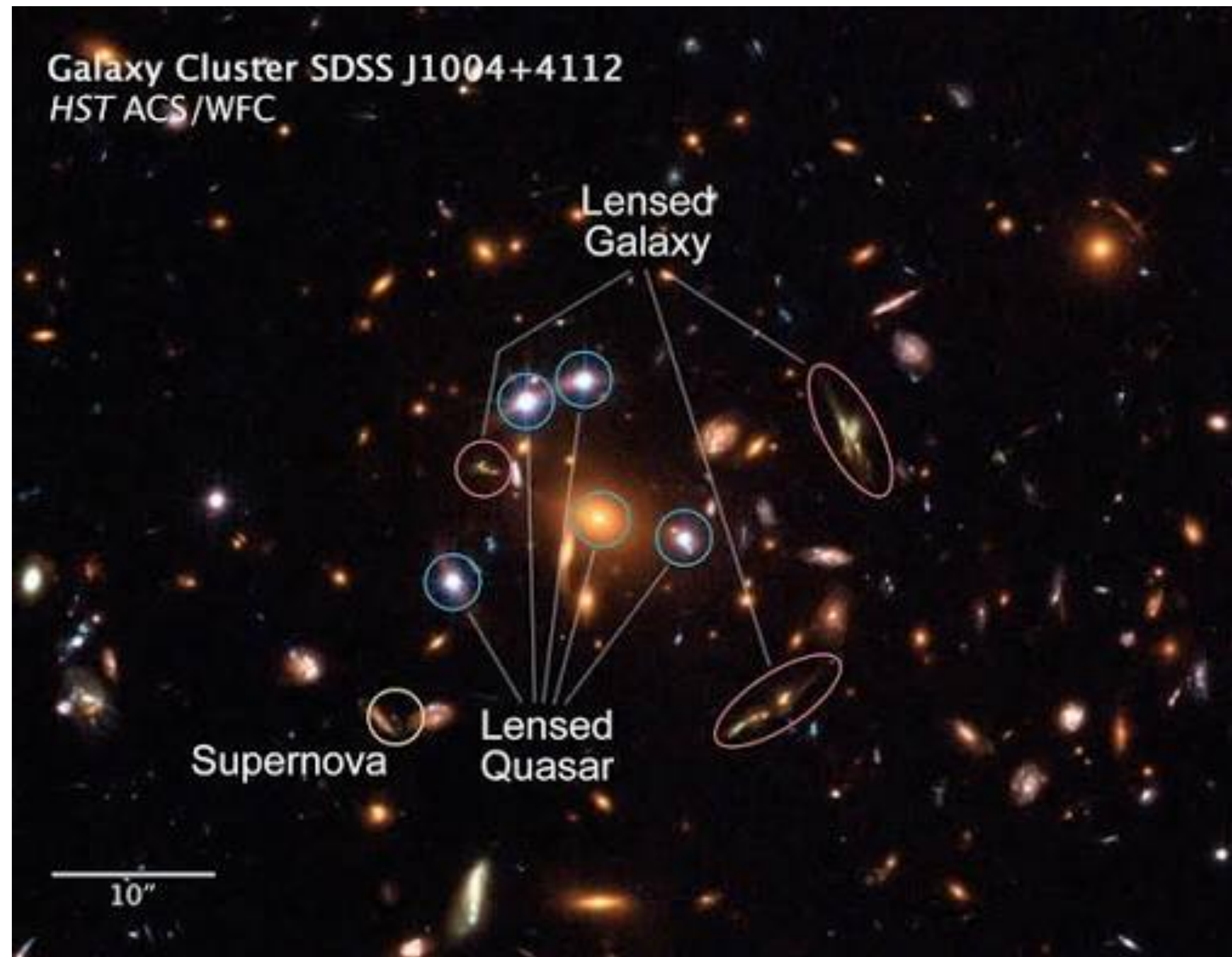
space is stretched: the throat is measured distance;
clocks run slow near the mass: $d\tau = \sqrt{1 - r_s/r} dt$ (GPS).



light cones tilt inward: at $r = r_s$ outgoing light runs in place; inside, all paths end at $r = 0$
(red: infalling, blue: outgoing light).

Gravity as a lens

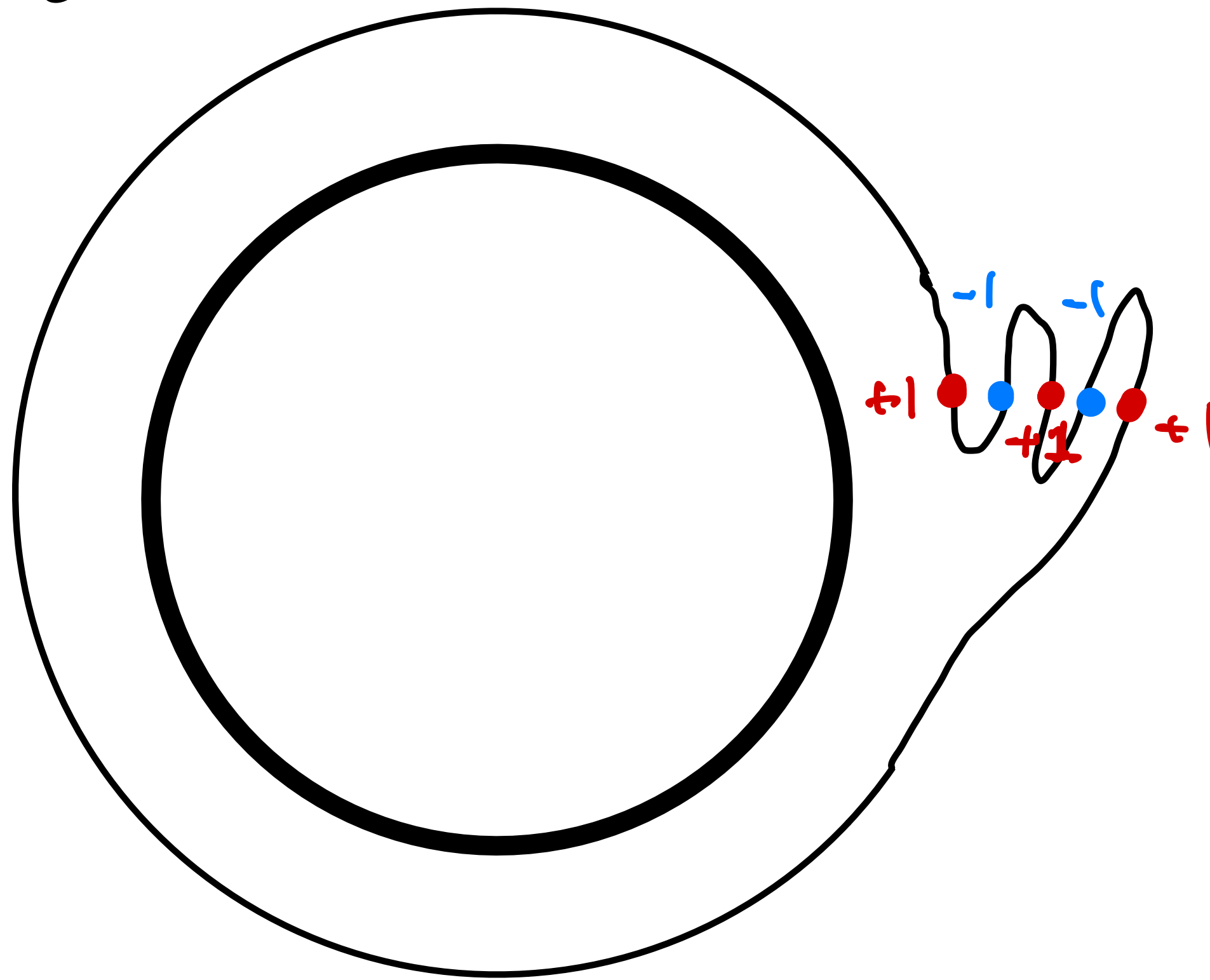




FACT. The number of gravitational images (if no image is blocked) is **always odd** — and **just less than half** of them are *inverted* (mirror) images.

Degree of a map

Net number of pre-images of a given point, counted with a “+” sign if the map is not inverted and a “−” sign if it is.



FACT. The number of gravitational images (if no image is blocked) is **always odd** — and **just less than half** of them are *inverted* (mirror) images.

SECTION 6

Summary and Outlook

Physics

springs & harmonic oscillators

monopoles & charge quantization

Biot–Savart law

quantum spin

time dilation & length contraction

gravity

gravitational lens

Mathematics

eigenvectors, Fourier analysis

Hopf bundle, Chern number

Gauss linking number, topology

vectors in $\mathbb{C}^2$, Hermitian matrices, $SU(2)$

Minkowski space, Lorentz transformation

curvature, geodesics

mapping degree, topology

- Spend time in the library and explore the wide range of physics and math books available
- Regularly browse mathematical magazines such as 「数学セミナー」 「数学のたのしみ」 「数理科学」 「数学」
- Go beyond your university courses by watching online lectures on topics that interest you, including those available on YouTube
- When you encounter something you do not understand, use AI as a tool to ask questions and clarify concepts.
- Most importantly, learn actively: work through problems yourself, write things down, and carry out calculations by hand.

あなたが気に入った方法，分野はきっとあなたに適しているのでしょう．ところで，あなたはすでに数学を志した．それは数学が好きになったからだ．数学のどんな所が好きになったのかよく反省して，数学のそのような所，方法，分野の発展した所を探せばよいかも知れない．しかし，数学は広いから，あなたのまだ知らぬ所に，まだまだあなたにピッタリのものがあるかも知れない．あまり早くから専門に固まらず，広い視野をもつことをお勧めします．

(久賀道郎／東京大学理学部数学科のオリエンテーションにて)

How simple are the laws of Nature?

Each fundamental theory is a single functional — the **action** S .

harmonic oscillator: $S = \int dt \left[\frac{m}{2} \dot{x}^2 - \frac{m\omega^2}{2} x^2 \right]$

free particle (geodesic): $S = -mc \int ds$

QED: $S = \int d^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi} i\gamma^\mu D_\mu \psi \right]$

QCD: $S = \int d^4x \left[-\frac{1}{2} \text{tr} F_{\mu\nu} F^{\mu\nu} + \bar{q} i\gamma^\mu D_\mu q \right]$

Einstein–Hilbert: $S = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda)$